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Table of Contents

1.0 Introduction.....	7
2.0 Entire Network Layout	9
3.0 IPv4 Addressing Table	10
4.0 LAN.....	11
4.1 VLAN Configuration.....	11
4.2 Voice VLAN	12
4.2.1 Johor.....	12
4.3 Access Port.....	14
4.3.1 KL	14
4.3.2 Penang.....	16
4.3.3 Johor.....	17
4.3.4 Server Farm	18
4.4 Trunk Port	19
4.4.1 KL	19
4.4.2 Penang.....	20
4.4.3 Johor.....	21
4.4.4 Server Farm	22
4.5 EtherChannel.....	23
4.5.1 KL	23
4.5.2 Johor.....	26
4.6 Router-on-a-stick.....	28
4.6.1 KL	28
4.6.2 Penang.....	29
4.6.3 Johor.....	30
4.6.4 Server Farm	31
4.7 HSRP (Hot Standby Router Protocol)	32
4.7.1 KL	32
4.7.2 Penang.....	33
4.7.3 Johor.....	34
4.7.4 Server Farm	35
5.0 WAN.....	36
5.1 OSPF (Open Shortest Path First)	36
5.1.1 KL	38
5.1.2 Penang.....	39
5.1.3 Johor.....	40

5.1.4 Server Farm	41
5.1.5 ISP.....	42
5.2 Default Static Route and Floating Static Route	43
5.2.1 KL	44
5.2.2 Penang.....	45
5.2.3 Johor.....	46
5.2.4 Server Farm	47
6.0 Penang WLAN	48
6.1 Overall Architecture Description	48
6.2 WLC Configuration.....	49
6.3 Verification of WLAN Configuration	54
7.0 Server Services in Server Farm.....	56
7.1 FTP Server	56
7.1.1 Configuration Process.....	56
7.1.2 Verification of FTP Server Configuration	58
7.2 Web Server.....	61
7.2.1 Configuration Process.....	61
7.2.2 Verification of Web Server Configuration.....	63
7.3 DNS Server	64
7.3.1 Configuration Process.....	64
7.3.2 Verification of DNS Server Configuration	66
7.4 DHCP Server.....	67
7.4.1 Configuration Process.....	67
7.4.2 Verification of DHCP Server Configuration.....	68
8.0 Tracert and Ping Result	69
8.1 Connection from KL to Server Farm.....	69
8.2 Connection from Johor to Penang	70
8.3 Connection from Penang to KL	71
8.4 Connection from Server Farm to Johor	72
8.5 Connection from KL to Johor	73
8.6 Connection from Penang to Server Farm	74
9.0 Basic Security Mechanisms	75
9.1 SSH.....	75
9.2 Layer 2 Attacks and Mitigation Techniques	77
9.2.1 First Layer 2 Attack.....	77
9.2.2 Second Layer 2 Attack.....	78
9.2.3 Third Layer 2 Attack.....	79

9.2.4 Fourth Layer 2 Attack.....	81
10.0 Recommendations to Improve the Security Mechanisms	83
11.0 Conclusion	84
12.0 References.....	85

Table of Figure

Figure 1 Entire Network Layout	9
Figure 2 IPv4 Addressing Table	10
Figure 3 KL Switch Figure 4 Penang Switch	11
Figure 5 Johor Switch Figure 6 Server Farm Switch	11
Figure 7 Voice VLAN	12
Figure 8 <show running-configuration>	13
Figure 9 Phone Dial	13
Figure 10 KL_Mgmt-Switch Access Port Figure 11 KL_HR-Switch Access Port ..	14
Figure 12 KL_Design-Switch Access Port.....	14
Figure 13 Penang Switch Access Port	16
Figure 14 JB_R&D-Switch Access Port	17
Figure 15 JB_Fin-Switch Access Port	17
Figure 16 SF-Switch Access Port	18
Figure 17 KL Switch Trunk Port	19
Figure 18 Penang Switch Trunk Port.....	20
Figure 19 Johor Switch Trunk Port.....	21
Figure 20 Server Farm Switch Server Farm.....	22
Figure 21 KL Port Channel	23
Figure 22 KL Switch <show etherchannel summary>	24
Figure 23 KL Switch <show interface trunk>	24
Figure 24 KL_HR-Switch <show etherchannel summary>	25
Figure 25 KL_HR-Switch <show interface trunk>	25
Figure 26 Johor Port Channel	26
Figure 27 Johor Switch <show etherchannel summary>	26
Figure 28 Johor Switch <show interface trunk>.....	26
Figure 29 JB_R&D-Switch <show etherchannel summary>	27
Figure 30 JB_R&D-Switch <show interface trunk>	27
Figure 31 KL Router 2	28
Figure 32 KL Router 1	28
Figure 33 PG Router 1	29
Figure 34 PG Router 2	29
Figure 35 JB Router 1	30
Figure 36 JB Router 2	30
Figure 37 SF Router 2	31
Figure 38 SF Router 1	31
Figure 39 KL Router 1 <show standby brief>	32
Figure 40 KL Router 2 <show standby brief>	32
Figure 41 PG Router 1 <show standby brief>	33
Figure 42 PG Router 2 <show standby brief>	33
Figure 43 JB Router 1 <show standby brief>	34
Figure 44 JB Router 2 <show standby brief>	34
Figure 45 SF Router 1 <show standby brief>	35
Figure 46 SF Router 2 <show standby brief>	35
Figure 47 OSPF Configuration Commands	36
Figure 48 KL_Router1 <show ip protocol>	38
Figure 49 KL_Router2 <show ip protocol>	38
Figure 50 PG_Router1 <show ip protocol>	39
Figure 51 PG_Router2 <show ip protocol>	39

Figure 52 JB_Router1 <show ip protocol>	40
Figure 53 JB_Router2 <show ip protocol>	40
Figure 54 SF_Router1 <show ip protocol>	41
Figure 55 SF_Router2 <show ip protocol>	41
Figure 56 ISP1 <show ip protocol>	42
Figure 57 ISP2 <show ip protocol>	42
Figure 58: Default and Floating Static Route Configurations	43
Figure 59 KL_Router1 <show running-config include ip route>	44
Figure 60 KL_Router2 <show running-config include ip route>	44
Figure 61 PG_Router1 <show running-config include ip route>	45
Figure 62 PG_Router2 <show running-config include ip route>	45
Figure 63 JB_Router1 <show running-config include ip route>	46
Figure 64 JB_Router2 <show running-config include ip route>	46
Figure 65 SF_Router1 <show running-config include ip route>	47
Figure 66 SF_Router2 <show running-config include ip route>	47
Figure 67 Penang WLAN	48
Figure 68 DHCP and IP Assignment	48
Figure 69 Interfaces Name	49
Figure 70 Interface Delivery.....	49
Figure 71 Interface Management	50
Figure 72 Interface IT.....	50
Figure 73 WLANs.....	51
Figure 74 WLANs PG_Delivery General	51
Figure 75 WLANs PG_Delivery Security.....	51
Figure 76 WLANs PG_Delivery PSK	52
Figure 77 AP Groups Name	53
Figure 78 Delivery_APGroup.....	53
Figure 79 PG_Delivery_AP LAP Registration.....	54
Figure 80 PG_Management_AP LAP Registration	54
Figure 81 PG_IT_AP LAP Registration	54
Figure 82 PG_Delivery-Laptop3 Configuration.....	55
Figure 83 PG_Delivery-Laptop3 Desktop	55
Figure 84 Johor Server Farm.....	56
Figure 85 IP Configuration.....	57
Figure 86 FTP Service.....	57

1.0 Introduction

This report documents the design and implementation of a secure, scalable, and efficient network infrastructure for Cloud Co, a networking company operating across three key sites: the headquarters in Kuala Lumpur (KL) and branch offices in Penang and Johor. The project was undertaken to modernize the existing network design, improve operational efficiency, and address critical security concerns, particularly in the headquarters. The report not only describes the logical network topology but also provides detailed justifications, configurations, and verification of the proposed design.

The primary objective of this project is to create a fully functional prototype network using Cisco Packet Tracer as the simulation tool. The design incorporates core networking elements such as VLAN segmentation, inter-VLAN routing, Voice VLAN integration, WAN routing strategies, WLAN deployment, and server configurations. Key technologies including EtherChannel, HSRP, and Router-on-a-Stick are configured to ensure redundancy, load balancing, and reliable communication between devices and sites. To support seamless connectivity across all locations, the network uses IPv4 addressing with VLSM, combining both private addressing for internal LAN and public addressing for WAN and servers.

In addition to connectivity, the network integrates a Server Farm hosting essential services such as FTP, Web, DNS, and DHCP. These servers are configured to ensure reliable access both within the local network and across different branches, supporting Cloud Co's operational requirements. A dedicated WLAN architecture is also implemented in the Penang branch, leveraging WLC-based management, VLAN-WLAN mapping, and WPA2 security protocols to provide secure and flexible wireless access.

Security plays a central role in this project. The documentation highlights Layer 2 attacks and their mitigations, focusing on common threats such as STP manipulation, BPDU flooding, and misuse of unused ports. Configurations for port security, blackhole VLANs, and other defensive measures are included to demonstrate practical mitigation strategies. Beyond these, the report also provides recommendations for further security enhancements, including advanced mechanisms such as honeypots and bastion hosts.

Overall, this report aims to demonstrate a comprehensive networking solution that not only fulfils Cloud Co s current operational needs but also ensures scalability, resilience, and security for future growth. The systematic documentation of configurations, verification commands, and security measures provides a clear understanding of how the network design aligns with both technical requirements and best practices in modern enterprise networking.

2.0 Entire Network Layout

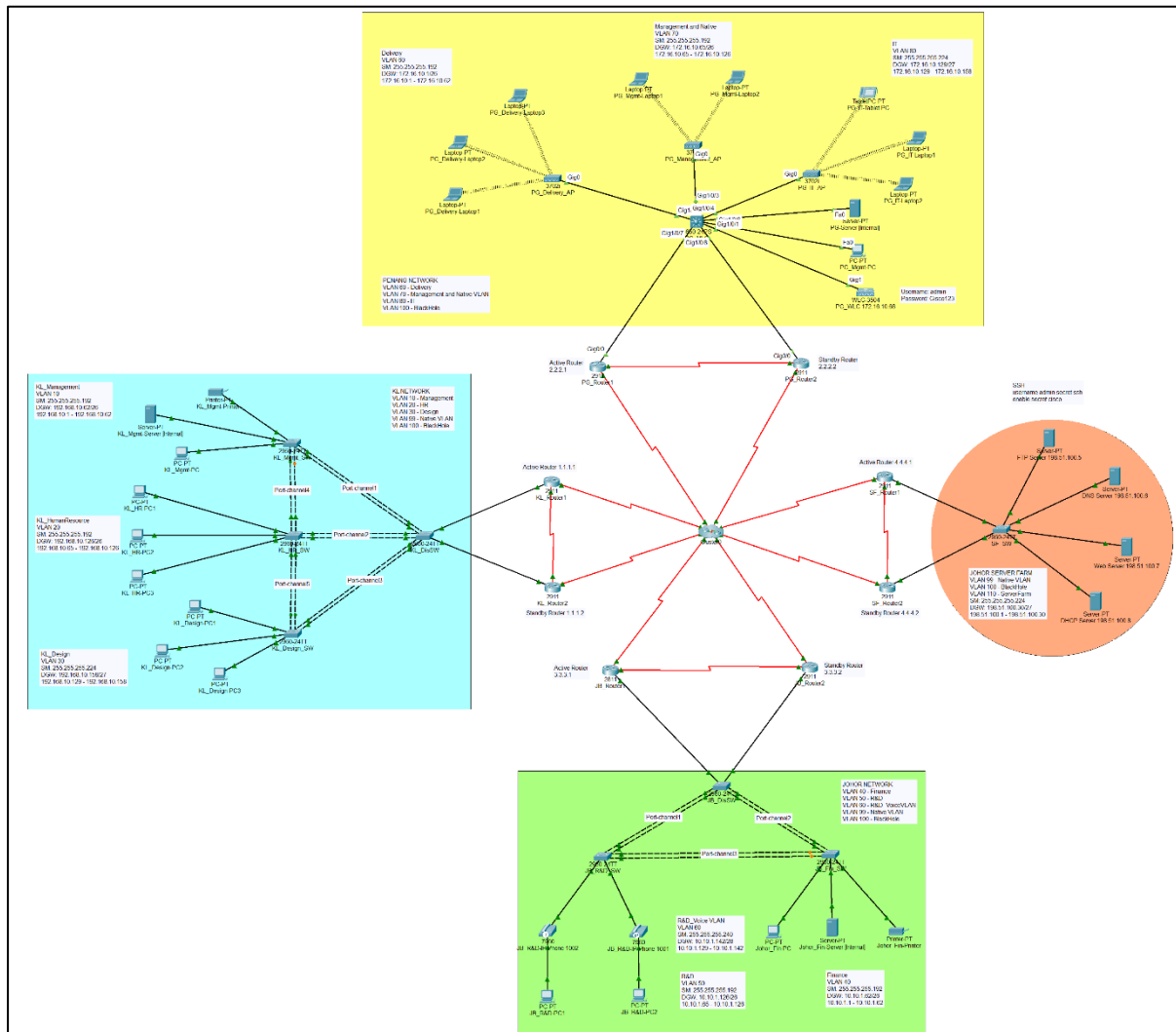


Figure 1 Entire Network Layout

The entire network layout integrates three major sites, which is KL (blue area), Penang (yellow area), and Johor (green area). The routers are configured with Router-on-a-Stick for inter-VLAN routing, combined with VLAN segmentation and HSRP redundancy to provide reliability of the gateways. Besides that, Johor Server Farm (orange area) will be offering centralized services like DHCP and DNS as well as Penang using a Wireless LAN Controller to handle the WLANs and LAPs. The trunk links with VLANs are set at the switches to facilitate the flow of communication within the departments. Lastly, security is strengthened through SSH access to network devices and WPA2-PSK as a connection to the wireless.

3.0 IPv4 Addressing Table

Cloud Co IPv4 Addressing Table						
KL Network	192.168.10.0/26					Private IP
Subnet	Block Size (Host - 0)	VLAN ID	Subnet Mask	Network Address	Range of Usable IP	Broadcast Address
KL_Management	50	10	255.255.255.192	192.168.10.0/26	192.168.10.1 - 192.168.10.62	192.168.10.63
KL_HumanResource	45	20	255.255.255.192	192.168.10.64/26	192.168.10.65 - 192.168.10.126	192.168.10.127
KL_Design	30	30	255.255.255.224	192.168.10.128/27	192.168.10.129 - 192.168.10.158	192.168.10.159
Native VLAN	16	99	255.255.255.240	192.168.10.160/28	192.168.10.161 - 192.168.10.174	192.168.10.175
BlackHole	Unused Port	100				
Johor Network	10.10.1.0/26					Private IP
Subnet	Block Size (Host - 0)	VLAN ID	Subnet Mask	Network Address	Range of Usable IP	Broadcast Address
Finance	50	40	255.255.255.192	10.10.1.0/26	10.10.1.1 - 10.10.1.62	10.10.1.63
R&D	35	50	255.255.255.192	10.10.1.64/26	10.10.1.65 - 10.10.1.126	10.10.1.127
R&D_Voice VLAN	15	60	255.255.255.240	10.10.1.128/28	10.10.1.129 - 10.10.1.142	10.10.1.143
Native VLAN	16	99	255.255.255.240	10.10.1.144/28	10.10.1.145 - 10.10.1.158	10.10.1.159
BlackHole	Unused Port	100				
Penang Network	172.16.10.0/26					Private IP
Subnet	Block Size (Host - 0)	VLAN ID	Subnet Mask	Network Address	Range of Usable IP	Broadcast Address
Delivery	50	60	255.255.255.192	172.16.10.0/26	172.16.10.1 - 172.16.10.62	172.16.10.63
Management	35	70	255.255.255.192	172.16.10.64/26	172.16.10.65 - 172.16.10.126	172.16.10.127
IT	25	80	255.255.255.224	172.16.10.128/27	172.16.10.129 - 172.16.10.158	172.16.10.159
Native VLAN		70				
BlackHole	Unused Port	100				
Server Farm (Johor)	198.51.100.0/27					Public IP
Subnet	Block Size (Host - 0)	VLAN ID	Subnet Mask	Network Address	Range of Usable IP	Broadcast Address
FTP, DNS, Web Server	20	110	255.255.255.224	198.51.100.0/27	198.51.100.1 - 198.51.100.30	198.51.100.31
Native VLAN	16	99	255.255.255.240	192.168.99.0/28	192.168.99.1 - 192.168.99.14	192.168.99.14
BlackHole	Unused Port	100				
WAN Connection	200.100.100.0/30					Public IP
Subnet	Block Size (Host - 0)	Subnet Mask	Network Address	Range of Usable IP	Broadcast Address	
WAN 1 (KL_Router1 > ISP1)	2	255.255.255.252	200.100.100.0/30	200.100.100.1 - 200.100.100.2	200.100.100.3	
WAN 2 (KL_Router2 > ISP2)	2	255.255.255.252	200.100.100.4/30	200.100.100.5 - 200.100.100.6	200.100.100.7	
WAN 3 (PG_Router1 > ISP1)	2	255.255.255.252	200.100.100.8/30	200.100.100.9 - 200.100.100.10	200.100.100.11	
WAN 4 (PG_Router2 > ISP2)	2	255.255.255.252	200.100.100.12/30	200.100.100.13 - 200.100.100.14	200.100.100.15	
WAN 5 (ISP1 > ISP2)	2	255.255.255.252	200.100.100.16/30	200.100.100.17 - 200.100.100.18	200.100.100.19	
WAN 6 (SF_Router1 > ISP2)	2	255.255.255.252	200.100.100.20/30	200.100.100.21 - 200.100.100.22	200.100.100.23	
WAN 7 (SF_Router2 > ISP1)	2	255.255.255.252	200.100.100.24/30	200.100.100.25 - 200.100.100.26	200.100.100.27	
WAN 8 (JB_Router1 > ISP2)	2	255.255.255.252	200.100.100.28/30	200.100.100.29 - 200.100.100.30	200.100.100.31	
WAN 9 (JB_Router2 > ISP1)	2	255.255.255.252	200.100.100.32/30	200.100.100.33 - 200.100.100.34	200.100.100.35	
Default and Floating Static Route	10.255.255.0/30					Private IP
Subnet	Block Size (Host - 0)	Subnet Mask	Network Address	Range of Usable IP	Broadcast Address	
KL_Router	2	255.255.255.252	10.255.255.0/30	10.255.255.1 - 10.255.255.2	10.255.255.3	
JB_Router	2	255.255.255.252	10.255.255.4/30	10.255.255.5 - 10.255.255.6	10.255.255.7	
SF_Router	2	255.255.255.252	10.255.255.8/30	10.255.255.9 - 10.255.255.10	10.255.255.11	
PG_Router	2	255.255.255.252	10.255.255.12/30	10.255.255.13 - 10.255.255.14	10.255.255.15	

Figure 2 IPv4 Addressing Table

VLSM was used to design the IPv4 addressing table to optimally use the IP address space and to support the needs of each subnet. The internal networks of KL, Penang and Johor were assigned private IP addresses range to facilitate communication between the departments. In addition, WAN connections and Johor Server Farm were implemented using public IP ranges so that they could be connected to the outside world and services. This hierarchical distribution improves scalability, management ease and security of communication throughout the network.

4.0 LAN

4.1 VLAN Configuration

All switches were given consistently names in terms of naming VLANs to enhance the network management and ease of use in the operations. Since each VLAN has a meaningful name, such as Management, Delivery, and IT, administrators can define the purpose of each VLAN at a glance. They do not need to use VLAN IDs. As a result, the practice will guarantee uniformity between the devices, minimize errors during configuration, and will simplify troubleshooting when using multiple switches.

```
KL_DisSW#sh vlan br
```

VLAN	Name	Status
1	default	active
10	Management	active
20	HumanResource	active
30	Design	active
99	Native	active
100	BlackHole	active
1002	fddi-default	active
1003	token-ring-default	active
1004	fddinet-default	active
1005	trnet-default	active

Figure 3 KL Switch

```
PG_MLS#sh vlan br
```

VLAN	Name	Status
1	default	active
60	Delivery	active
70	Management_and_Native	active
80	IT	active
100	BlackHole	active
1002	fddi-default	active
1003	token-ring-default	active
1004	fddinet-default	active
1005	trnet-default	active

Figure 4 Penang Switch

```
JB_DisSW#sh vlan br
```

VLAN	Name	Status
1	default	active
40	Finance	active
50	R&D	active
60	Voice	active
99	Native	active
100	BlackHole	active
1002	fddi-default	active
1003	token-ring-default	active
1004	fddinet-default	active
1005	trnet-default	active

Figure 5 Johor Switch

```
SF_SW#sh vlan br
```

VLAN	Name	Status
1	default	active
99	Native	active
100	BlackHole	active
110	ServerFarm	active
1002	fddi-default	active
1003	token-ring-default	active
1004	fddinet-default	active
1005	trnet-default	active

Figure 6 Server Farm Switch

4.2 Voice VLAN

The voice VLAN is a special port that was purposely designed to carry voice over IP traffic. It is because voice needs low delay and clear transmission when they are transmitting as it keeps both data and voice traffic separate. Besides, voice VLAN is more convenient to apply rules, monitoring or policies.

4.2.1 Johor

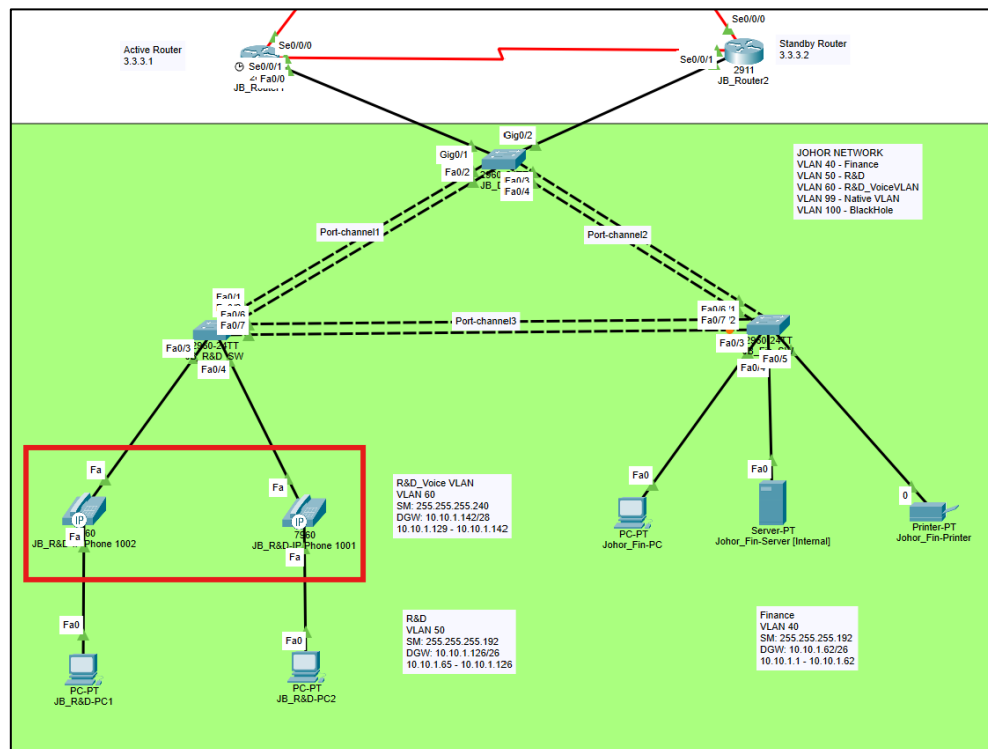


Figure 7 Voice VLAN

The IP phone was designed in Johor Network to ensure their communication. Even though PC and IP phone are plugged in the same port, they are separate to two different VLAN since they use different traffic which is voice traffic for IP Phone, and data traffic for the PC. Hence, the IP phone finds the source from VLAN 60 which brings a different IP address. This guarantees the quality of the call and keeps voice and data traffic apart. The IP phone designed allows them to call each other. Once they reach, it proves that the IP phone works smoothly in the cisco file.

```
JB_Router1#show running-config | section telephony-service
telephony-service
max-ephones 5
max-dn 5
ip source-address 10.10.1.129 port 2000
auto assign 1 to 2
```

Figure 8 <show running-configuration>

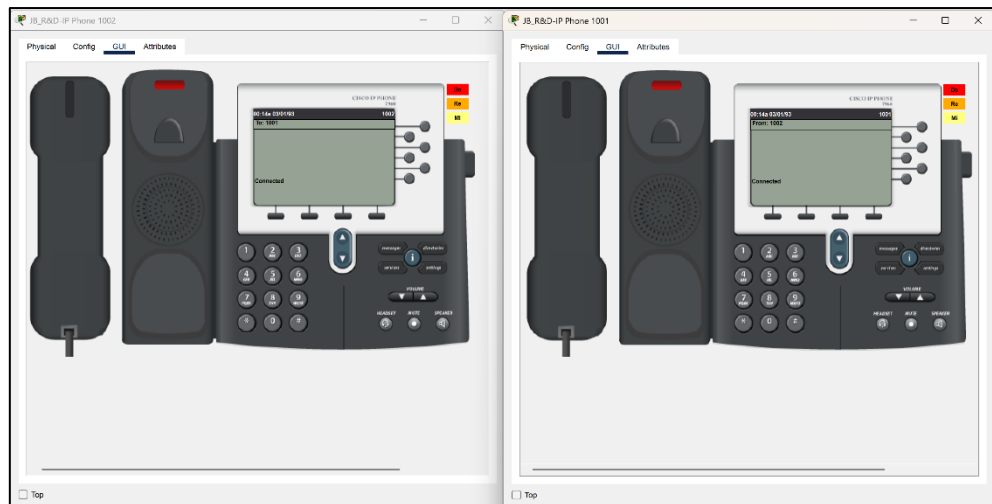


Figure 9 Phone Dial

4.3 Access Port

Access ports are switch interfaces that connect directly to end-user devices such as computers, printers, and IP phones. These ports are configured to carry traffic for only one VLAN, ensuring that devices in different departments remain properly segmented. This prevents unnecessary cross-communication and enforces network security.

4.3.1 KL

KL_Mgmt-Switch

```
KL_Mgmt-Switch#sh running-conf
Building configuration...

interface FastEthernet0/11
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/12
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/13
 switchport access vlan 10
 switchport mode access
!
```

Figure 10 KL_Mgmt-Switch Access Port

KL_HR-Switch

```
KL_HR-Switch#sh running-config
Building configuration...

interface FastEthernet0/11
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/12
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/13
 switchport access vlan 20
 switchport mode access
!
```

Figure 11 KL_HR-Switch Access Port

KL_Design-Switch

```
KL_Design-Switch#sh running-conf
Building configuration...

interface FastEthernet0/11
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/12
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/13
 switchport access vlan 30
 switchport mode access
!
```

Figure 12 KL_Design-Switch Access Port

In KL, access ports are configured on departmental switches to provide connectivity for end-user devices. On the KL_Mgmt-Switch, ports Fa0/11–

Fa0/13 are set as access mode and assigned to VLAN 10 for the Management department. This ensures that all management devices connect to the correct network segment. Similarly, the KL_HR-Switch assigns ports to VLAN 20 to isolate Human Resources devices, protecting sensitive employee data. The KL_Design-Switch is assigned to VLAN 30, which supports design-specific applications and higher bandwidth allocation.

4.3.2 Penang

```
PG_MLS#sh running-config
Building configuration...

interface GigabitEthernet1/0/2
 switchport access vlan 60
 switchport mode access
!
interface GigabitEthernet1/0/3
 switchport access vlan 70
 switchport mode access
!
interface GigabitEthernet1/0/4
 switchport access vlan 80
 switchport mode access
 shutdown
!
interface GigabitEthernet1/0/5
 switchport access vlan 70
 switchport mode access
!
```

Figure 13 Penang Switch Access Port

In Penang, access ports are assigned according to departmental requirements. On the switch PG_MLS, the configuration shows that Gig1/0/2 is set to access mode and assigned to VLAN 60 for the Delivery department. The Management department is connected through Gig1/0/3 and Gig1/0/5, both configured as access ports in VLAN 70. The IT department is assigned to Gig1/0/4, which is configured for VLAN 80.

By assigning specific ports to each VLAN, departmental devices are placed into their designated network segments. This ensures proper traffic separation, while inter-VLAN routes allows necessary communication between departments when required.

4.3.3 Johor

R&D Department :

```
JB_R&D-Switch#sh running-config
Building configuration...

Current configuration : 1333 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname JB_R&D-Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
 switchport access vlan 50
 switchport mode access
 switchport voice vlan 60
!
interface FastEthernet0/4
 switchport access vlan 50
 switchport mode access
 switchport voice vlan 60
```

Figure 14 JB_R&D-Switch Access Port

Finance Department :

```
JB_Fin-Switch#sh running-config
Building configuration...

Current configuration : 1332 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname JB_Fin-Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 40
 switchport mode access
```

Figure 15 JB_Fin-Switch Access Port

In Johor, access ports are assigned to support both the R&D and Finance departments, including IP telephony. On the JB_R&D_SW, Fa0/3 are configured for VLAN 50 that is R&D, while Fa0/4 is configured for VLAN 60 that is R&D_Voice to provide IP phone connectivity alongside computer data. This dual setup enables R&D devices to use the same cabling infrastructure for both PCs and IP phones while maintaining logical separation.

On the JB_Fin_SW, finance department devices are connected to Fa0/3 - Fa0/5, which are configured as access ports for VLAN 40. This isolates the Finance department's devices, including the Finance server and printer, from other VLANs to maintain the confidentiality and security of financial data.

4.3.4 Server Farm

```
SF_Switch#sh running-config
Building configuration...

Current configuration : 1386 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname SF_Switch
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 110
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 110
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 110
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 110
 switchport mode access
!
```

Figure 16 SF-Switch Access Port

In the Server Farm, access ports are dedicated to critical infrastructure services. Fa0/1 – Fa0/4 are configured as access ports assigned to VLAN 110 (Server Farm). These ports connect servers such as DNS, DHCP, FTP, and Web servers, ensuring that all core services run in a dedicated VLAN separate from user traffic.

This design not only prevents congestion by isolating server traffic from end-user VLANs but also allows administrators to apply stricter security and monitoring policies to protect these critical systems. The VLAN segregation in the Server Farm ensures high availability, optimized performance, and controlled access to vital network services.

4.4 Trunk Port

Unlike access ports, trunk ports are designed to carry traffic for multiple VLANs across a single physical link. This is crucial when interconnecting switches or connecting switches to routers for inter-VLAN routing. Trunk ports reduce the number of required cables while still maintaining VLAN separation.

4.4.1 KL

```
KL_DisSW#sh int tr
```

Port	Mode	Encapsulation	Status	Native vlan
Po1	on	802.1q	trunking	99
Po2	on	802.1q	trunking	99
Po3	on	802.1q	trunking	99
Gig0/1	on	802.1q	trunking	99
Gig0/2	on	802.1q	trunking	99

Port	Vlans allowed on trunk
Po1	10,20,30,99
Po2	10,20,30,99
Po3	10,20,30,99
Gig0/1	10,20,30,99
Gig0/2	10,20,30,99

Port	Vlans allowed and active in management domain
Po1	10,20,30,99
Po2	10,20,30,99
Po3	10,20,30,99
Gig0/1	10,20,30,99
Gig0/2	10,20,30,99

Port	Vlans in spanning tree forwarding state and not pruned
Po1	10,20,30,99
Po2	10,20,30,99
Po3	10,20,30,99
Gig0/1	10,20,30,99
Gig0/2	10,20,30,99

Figure 17 KL Switch Trunk Port

In KL, a trunk port is configured on the distribution switch to connect to the router. The interface Fa0/1 is set as trunk mode, with VLAN 99 assigned as the native VLAN. Allowed VLANs include 10, 20, 30, and 99, ensuring traffic from the Management, HR, and Design departments is carried across the trunk. Using VLAN 99 instead of the default VLAN 1 improves security against VLAN hopping.

4.4.2 Penang

PG_MLS#sh int tr				
Port	Mode	Encapsulation	Status	Native vlan
Gig1/0/1	on	802.1q	trunking	70
Gig1/0/2	on	802.1q	trunking	70
Gig1/0/3	on	802.1q	trunking	70
Gig1/0/4	on	802.1q	trunking	70
Gig1/0/7	on	802.1q	trunking	70
Gig1/0/8	on	802.1q	trunking	70
Port	Vlans allowed on trunk			
Gig1/0/1	60,70,80			
Gig1/0/2	60,70,80			
Gig1/0/3	60,70,80			
Gig1/0/4	60,70,80			
Gig1/0/7	60,70,80			
Gig1/0/8	60,70,80			
Port	Vlans allowed and active in management domain			
Gig1/0/1	60,70,80			
Gig1/0/2	60,70,80			
Gig1/0/3	60,70,80			
Gig1/0/4	60,70,80			
Gig1/0/7	60,70,80			
Gig1/0/8	60,70,80			
Port	Vlans in spanning tree forwarding state and not pruned			
Gig1/0/1	60,70,80			
Gig1/0/2	60,70,80			
Gig1/0/3	60,70,80			
Gig1/0/4	60,70,80			
Gig1/0/7	60,70,80			
Gig1/0/8	60,70,80			

Figure 18 Penang Switch Trunk Port

In Penang, trunk ports are configured on the multilayer switch to interconnect the departmental switches. The output shows that interfaces Gig1/0/1, Gig1/0/2, Gig1/0/3, Gig1/0/4, Gig1/0/7, and Gig1/0/8 are operating in trunk mode with 802.1Q encapsulation, and VLAN 70 is set as the native VLAN. These trunk links allow VLAN 60 (Delivery), VLAN 70 (Management), and VLAN 80 (IT) to be carried across the network, ensuring that departmental VLAN traffic is properly transmitted while maintaining separation on the same physical connections.

4.4.3 Johor

JB_DisSW#sh int tr				
Port	Mode	Encapsulation	Status	Native vlan
Pol	on	802.1q	trunking	99
Po2	on	802.1q	trunking	99
Gig0/1	on	802.1q	trunking	99
Gig0/2	on	802.1q	trunking	99
Port	Vlans allowed on trunk			
Pol	40,50,60,99			
Po2	40,50,60,99			
Gig0/1	40,50,60,99			
Gig0/2	40,50,60,99			
Port	Vlans allowed and active in management domain			
Pol	40,50,60,99			
Po2	40,50,60,99			
Gig0/1	40,50,60,99			
Gig0/2	40,50,60,99			
Port	Vlans in spanning tree forwarding state and not pruned			
Pol	40,50,60,99			
Po2	40,50,60,99			
Gig0/1	40,50,60,99			
Gig0/2	40,50,60,99			

Figure 19 Johor Switch Trunk Port

In Johor, trunk ports are configured on the distribution switch to support departmental VLAN traffic. The output shows that interfaces Fa0/1 and Fa0/2 are operating in trunk mode with 802.1Q encapsulation, and VLAN 99 is used as the native VLAN. These trunks carry VLAN 40 (Finance), VLAN 50 (R&D data), and VLAN 60 (Voice), ensuring both data and VoIP traffic are properly transmitted with VLAN separation and QoS prioritization for voice traffic.

4.4.4 Server Farm

SF_SW#sh int tr				
Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	99
Gig0/2	on	802.1q	trunking	99
Port	Vlans allowed on trunk			
Gig0/1	99,110			
Gig0/2	99,110			
Port	Vlans allowed and active in management domain			
Gig0/1	99,110			
Gig0/2	99,110			
Port	Vlans in spanning tree forwarding state and not pruned			
Gig0/1	99,110			
Gig0/2	99,110			

Figure 20 Server Farm Switch Server Farm

In the Server Farm, trunk ports are configured to handle management and server traffic. The output shows that GigabitEthernet0/1 is operating as a trunk with 802.1Q encapsulation, and VLAN 99 is set as the native VLAN. The trunk link only allows VLAN 70 (Management) and VLAN 110 (Server Farm), ensuring critical server traffic is isolated from user VLANs while maintaining secure management access.

4.5 EtherChannel

EtherChannel technology allows multiple physical links to be bundled into a single logical link. This provides higher bandwidth, redundancy, and automatic failovers while simplifying spanning tree protocol operation.

4.5.1 KL

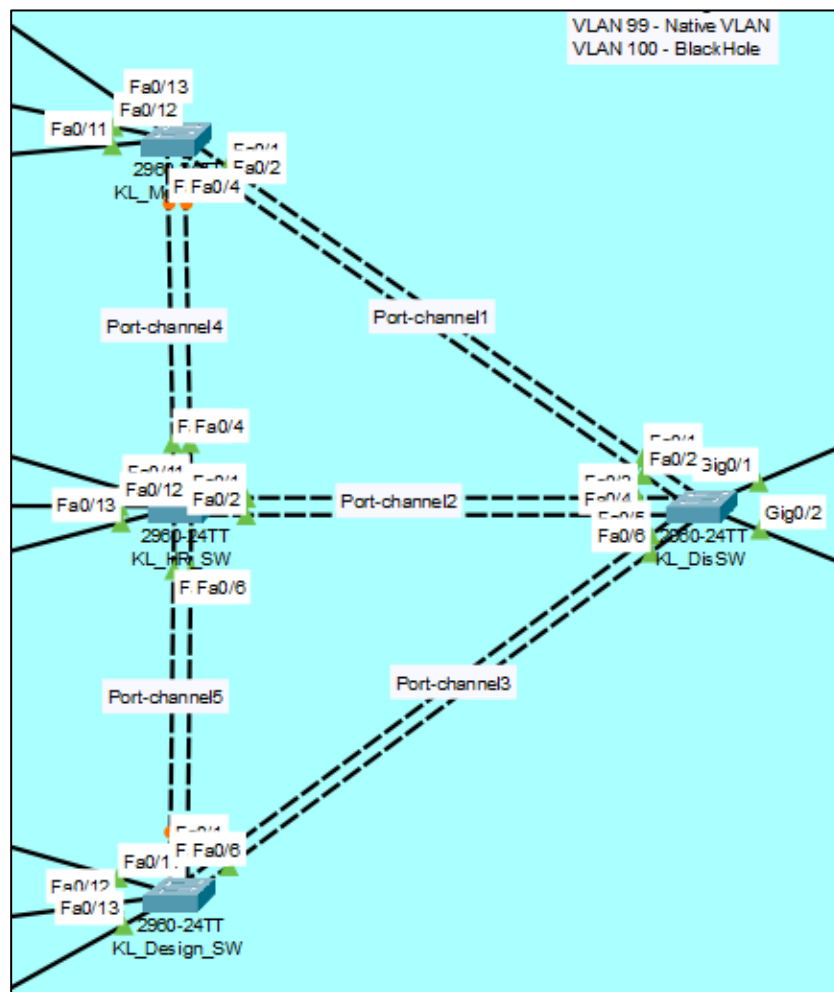


Figure 21 KL Port Channel

```

KL_DisSW(config)#do sh etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 3
Number of aggregators:          3

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SU)          LACP        Fa0/1 (P) Fa0/2 (P)
2      Po2 (SU)          LACP        Fa0/3 (P) Fa0/4 (P)
3      Po3 (SU)          LACP        Fa0/5 (P) Fa0/6 (P)

```

Figure 22 KL Switch <show etherchannel summary>

```

KL_DisSW#sh int tr
Port      Mode      Encapsulation  Status      Native vlan
Po1       on        802.1q         trunking    99
Po2       on        802.1q         trunking    99
Po3       on        802.1q         trunking    99
Gig0/1    on        802.1q         trunking    99
Gig0/2    on        802.1q         trunking    99

Port      Vlans allowed on trunk
Po1       10,20,30,99
Po2       10,20,30,99
Po3       10,20,30,99
Gig0/1    10,20,30,99
Gig0/2    10,20,30,99

```

Figure 23 KL Switch <show interface trunk>

In KL, EtherChannel is implemented on the distribution switch to provide redundant and high-speed connectivity between the access switches and the core. Multiple FastEthernet interfaces are bundled into logical links, forming Port-Channel1, Port-Channel2, and Port-Channel3. Each Port-Channel connects the KL_Mgmt-SW, KL_HR-SW, and KL_Design-SW respectively to the distribution switch. The configuration uses LACP in active mode, which allows automatic negotiation of channel formation and continuous monitoring of link health.


```

KL_HR_SW#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 3
Number of aggregators:          3

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
2      Po2 (SU)          LACP        Fa0/1 (P) Fa0/2 (P)
4      Po4 (SU)          LACP        Fa0/3 (P) Fa0/4 (P)
5      Po5 (SU)          LACP        Fa0/5 (P) Fa0/6 (P)

```

Figure 24 KL_HR-Switch <show etherchannel summary>

```

KL_HR_SW#sh int tr
Port    Mode        Encapsulation  Status      Native vlan
Po2     on          802.1q         trunking    99
Po4     on          802.1q         trunking    99
Po5     on          802.1q         trunking    99

Port      Vlans allowed on trunk
Po2       10,20,30,99
Po4       10,20,30,99
Po5       10,20,30,99

```

Figure 25 KL_HR-Switch <show interface trunk>

Additionally, Port-Channel4 and Port-Channel5 are created as lateral EtherChannels between the access switches themselves (KL_Mgmt-SW ↔ KL_HR-SW, and KL_HR-SW ↔ KL_Design-SW). These links are configured as trunks carrying VLANs 10 (Management), 20 (HR), 30 (Design), and 99 (Native). By introducing these cross-connections, the design not only increases available bandwidth but also adds redundancy in case the distribution switch becomes a single point of failure.

4.5.2 Johor

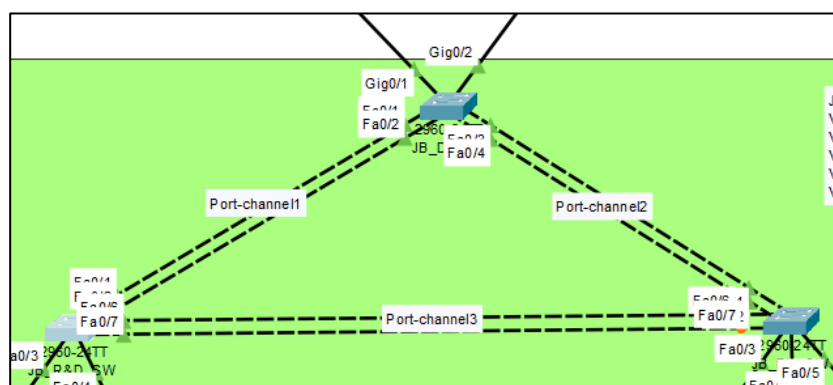


Figure 26 Johor Port Channel

```
JB_DisSW(config-if)#do sh etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 2
Number of aggregators:          2

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SU)          LACP        Fa0/1 (P) Fa0/2 (P)
2      Po2 (SU)          LACP        Fa0/3 (P) Fa0/4 (P)
```

Figure 27 Johor Switch <show etherchannel summary>

```
JB_DisSW#sh int tr
Port      Mode          Encapsulation  Status      Native vlan
Po1       on            802.1q         trunking    99
Po2       on            802.1q         trunking    99
Gig0/1    on            802.1q         trunking    99
Gig0/2    on            802.1q         trunking    99

Port      Vlans allowed on trunk
Po1       40,50,60,99
Po2       40,50,60,99
Gig0/1    40,50,60,99
Gig0/2    40,50,60,99
```

Figure 28 Johor Switch <show interface trunk>

In Johor, EtherChannel technology is deployed through three logical Port-Channels to ensure redundancy, high bandwidth, and reliable departmental connectivity. Port-Channel1 connects the Johor distribution switch to the upstream core network, providing a resilient and aggregated uplink path. Port-Channel2 links the distribution switch with the Finance access switch, carrying VLAN 40 for financial systems along with the native VLAN 99.

```

JB_R&D_SW#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  S - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 2
Number of aggregators:          2

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SU)      LACP        Fa0/1 (P) Fa0/2 (P)
3      Po3 (SU)      LACP        Fa0/6 (P) Fa0/7 (P)

```

Figure 29 JB_R&D-Switch <show etherchannel summary>

```

JB_R&D_SW#sh int tr
Port      Mode          Encapsulation  Status      Native vlan
Po1       on             802.1q         trunking    99
Po3       on             802.1q         trunking    99

Port      Vlans allowed on trunk
Po1       40,50,60,99
Po3       40,50,60,99

```

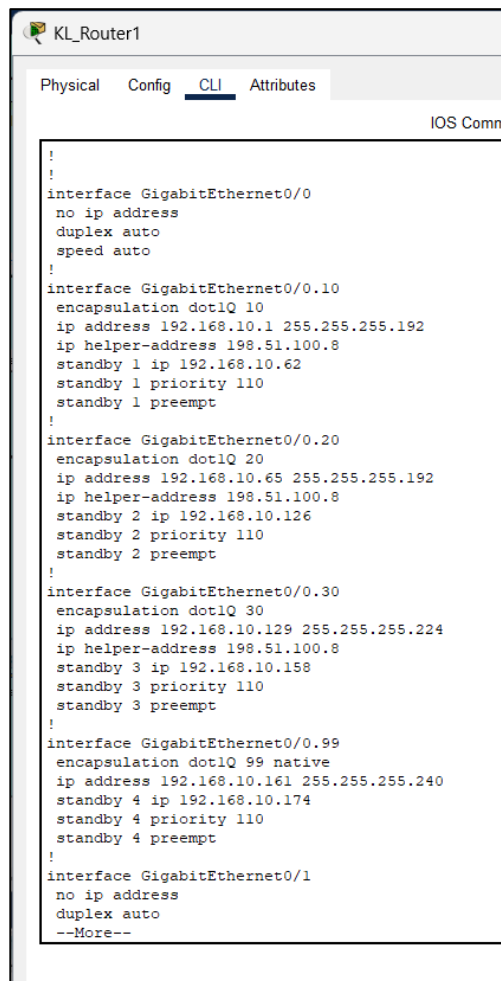
Figure 30 JB_R&D-Switch <show interface trunk>

Port-Channel3 connects the distribution switch with the R&D access switch, transporting VLAN 50 (R&D data) and VLAN 60 (R&D Voice), in addition to VLAN 99 for native traffic.

By aggregating multiple physical FastEthernet interfaces into each Port-Channel, the design doubles the available bandwidth while ensuring that a single link failure does not disrupt communication. VLAN 60, configured as a Voice VLAN, benefits from QoS prioritization, ensuring clear IP telephony performance for the R&D department.

4.6 Router-on-a-stick

4.6.1 KL



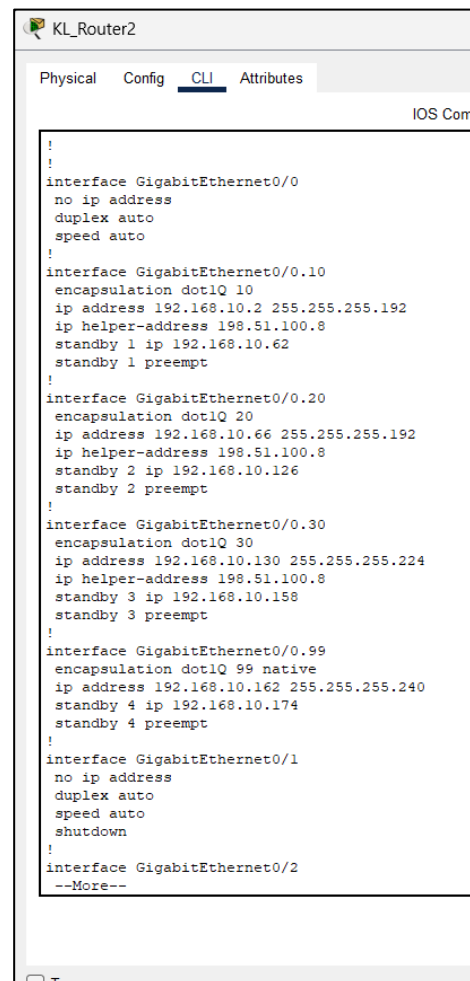
The screenshot shows the CLI configuration for KL_Router1. The configuration is as follows:

```

!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.192
ip helper-address 198.51.100.8
standby 1 ip 192.168.10.62
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.10.65 255.255.255.192
ip helper-address 198.51.100.8
standby 2 ip 192.168.10.126
standby 2 priority 110
standby 2 preempt
!
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 192.168.10.129 255.255.255.224
ip helper-address 198.51.100.8
standby 3 ip 192.168.10.158
standby 3 priority 110
standby 3 preempt
!
interface GigabitEthernet0/0.99
encapsulation dot1Q 99 native
ip address 192.168.10.161 255.255.255.240
standby 4 ip 192.168.10.174
standby 4 priority 110
standby 4 preempt
!
interface GigabitEthernet0/1
no ip address
duplex auto
--More--

```

Figure 32 KL Router 1



The screenshot shows the CLI configuration for KL_Router2. The configuration is as follows:

```

!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.2 255.255.255.192
ip helper-address 198.51.100.8
standby 1 ip 192.168.10.62
standby 1 preempt
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.10.66 255.255.255.192
ip helper-address 198.51.100.8
standby 2 ip 192.168.10.126
standby 2 preempt
!
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 192.168.10.130 255.255.255.224
ip helper-address 198.51.100.8
standby 3 ip 192.168.10.158
standby 3 preempt
!
interface GigabitEthernet0/0.99
encapsulation dot1Q 99 native
ip address 192.168.10.162 255.255.255.240
standby 4 ip 192.168.10.174
standby 4 preempt
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/2
--More--

```

Figure 31 KL Router 2

In KL, there are two router which is an active router (KL_Router1) and a standby router (KL_Router2). As the figure shows that, the three sub interfaces: G0/0/0.10, G0/0/0.20, and G0/0/0.30 are assigned for KL management, human resource, and design department VLANs. Moreover, the sub interface G0/0/0.99 is encapsulated for native VLAN with dot1Q native.

4.6.2 Penang

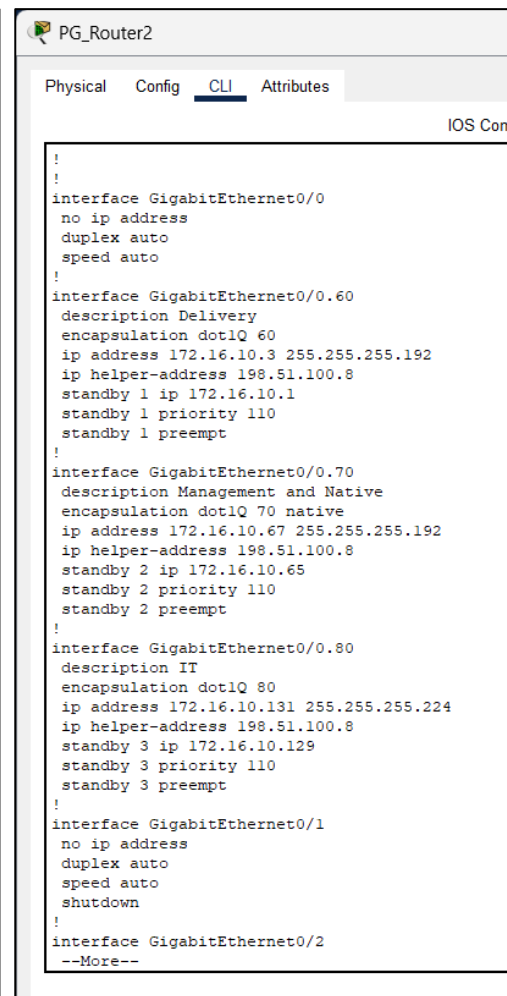


```

PG_Router1
Physical Config CLI Attributes
IOS
!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.60
description Delivery
encapsulation dot1Q 60
ip address 172.16.10.2 255.255.255.192
ip helper-address 198.51.100.8
standby 1 ip 172.16.10.1
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/0.70
description Management and Native
encapsulation dot1Q 70 native
ip address 172.16.10.66 255.255.255.192
ip helper-address 198.51.100.8
standby 2 ip 172.16.10.65
standby 2 priority 110
standby 2 preempt
!
interface GigabitEthernet0/0.80
description IT
encapsulation dot1Q 80
ip address 172.16.10.130 255.255.255.224
ip helper-address 198.51.100.8
standby 3 ip 172.16.10.129
standby 3 priority 110
standby 3 preempt
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/2
--More--

```

Figure 33 PG Router 1



```

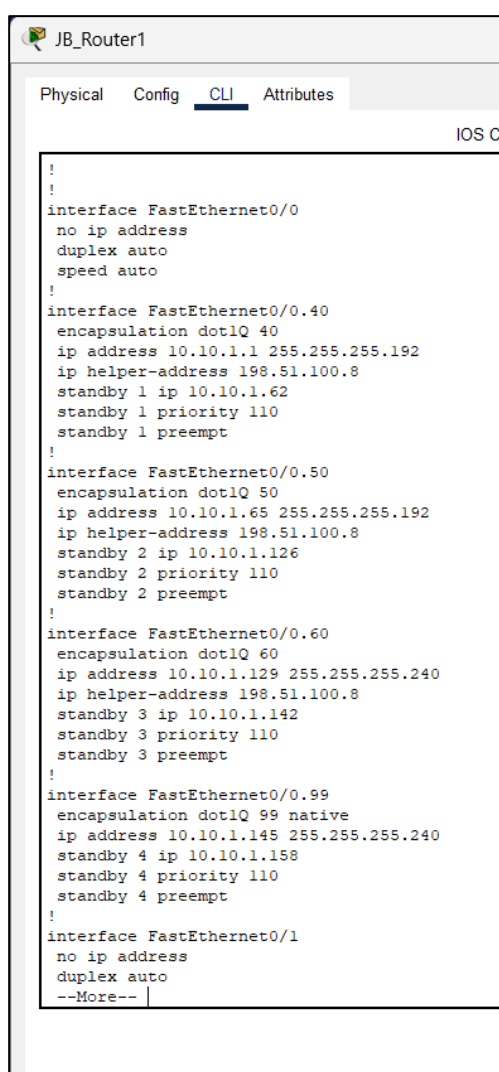
PG_Router2
Physical Config CLI Attributes
IOS Conf
!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.60
description Delivery
encapsulation dot1Q 60
ip address 172.16.10.3 255.255.255.192
ip helper-address 198.51.100.8
standby 1 ip 172.16.10.1
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/0.70
description Management and Native
encapsulation dot1Q 70 native
ip address 172.16.10.67 255.255.255.192
ip helper-address 198.51.100.8
standby 2 ip 172.16.10.65
standby 2 priority 110
standby 2 preempt
!
interface GigabitEthernet0/0.80
description IT
encapsulation dot1Q 80
ip address 172.16.10.131 255.255.255.224
ip helper-address 198.51.100.8
standby 3 ip 172.16.10.129
standby 3 priority 110
standby 3 preempt
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/2
--More--

```

Figure 34 PG Router 2

In Penang branch, there are two router which is an active router (PG_Router1) and a standby router (PG_Router2). As the figure shows that, the sub interfaces G0/0/0.60 and G0/0/0.80 are configured for Penang delivery and IT department. Moreover, the sub interface G0/0/0.70 is configured for management and encapsulated for native VLAN with dot1Q native.

4.6.3 Johor



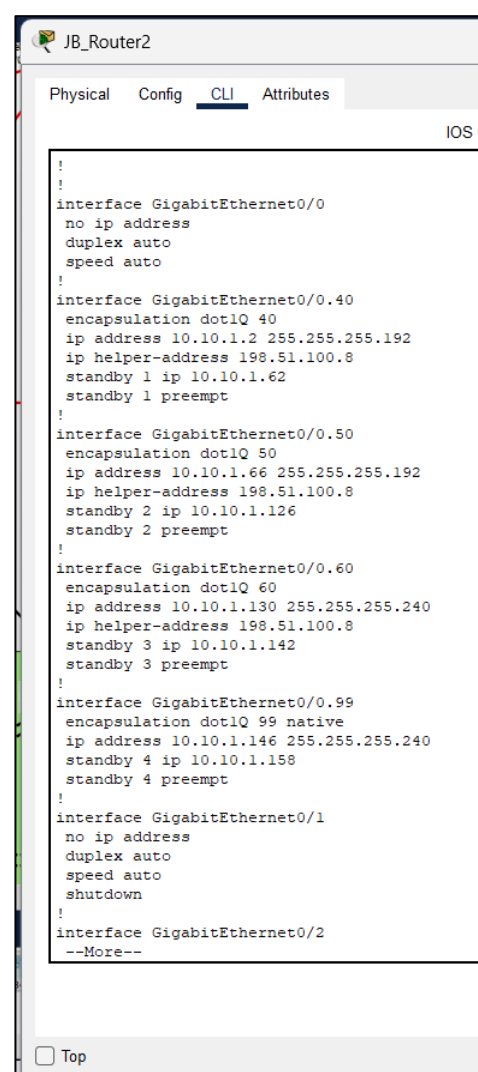
```

JB_Router1
Physical Config CLI Attributes
IOS C

!
!
interface FastEthernet0/0
no ip address
duplex auto
speed auto
!
interface FastEthernet0/0.40
encapsulation dot1Q 40
ip address 10.10.1.1 255.255.255.192
ip helper-address 198.51.100.8
standby 1 ip 10.10.1.62
standby 1 priority 110
standby 1 preempt
!
interface FastEthernet0/0.50
encapsulation dot1Q 50
ip address 10.10.1.65 255.255.255.192
ip helper-address 198.51.100.8
standby 2 ip 10.10.1.126
standby 2 priority 110
standby 2 preempt
!
interface FastEthernet0/0.60
encapsulation dot1Q 60
ip address 10.10.1.129 255.255.255.240
ip helper-address 198.51.100.8
standby 3 ip 10.10.1.142
standby 3 priority 110
standby 3 preempt
!
interface FastEthernet0/0.99
encapsulation dot1Q 99 native
ip address 10.10.1.145 255.255.255.240
standby 4 ip 10.10.1.158
standby 4 priority 110
standby 4 preempt
!
interface FastEthernet0/1
no ip address
duplex auto
--More--

```

Figure 35 JB Router 1



```

JB_Router2
Physical Config CLI Attributes
IOS C

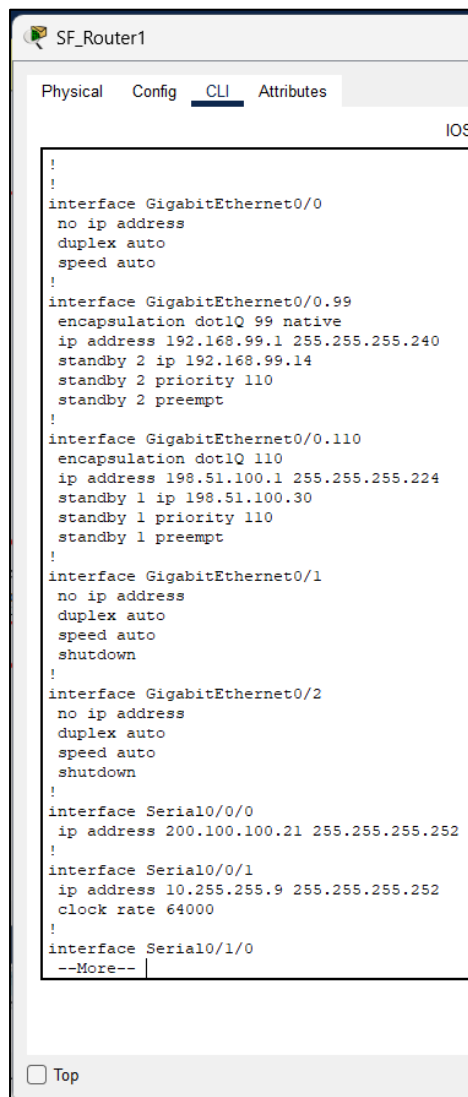
!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 10.10.1.2 255.255.255.192
ip helper-address 198.51.100.8
standby 1 ip 10.10.1.62
standby 1 preempt
!
interface GigabitEthernet0/0.50
encapsulation dot1Q 50
ip address 10.10.1.66 255.255.255.192
ip helper-address 198.51.100.8
standby 2 ip 10.10.1.126
standby 2 preempt
!
interface GigabitEthernet0/0.60
encapsulation dot1Q 60
ip address 10.10.1.130 255.255.255.240
ip helper-address 198.51.100.8
standby 3 ip 10.10.1.142
standby 3 preempt
!
interface GigabitEthernet0/0.99
encapsulation dot1Q 99 native
ip address 10.10.1.146 255.255.255.240
standby 4 ip 10.10.1.158
standby 4 preempt
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/2
--More--

```

Figure 36 JB Router 2

In Johor branch, there are two router which is an active router (JB_Router1) and a standby router (JB_Router2). As the figure shows that, the three sub interfaces in active router: Fa0/0/0.40, Fa0/0/0.50, and Fa0/0/0.60 are assigned for JB Finance, R&D department VLANs and R&D Voice VLAN. In standby router, G0/0/0.40, G0/0/0.50, and G0/0/0.60 are assigned for JB Finance, R&D department VLANs and R&D Voice VLAN. Moreover, the sub interface G0/0/0.99 is encapsulated for native VLAN with dot1Q native.

4.6.4 Server Farm



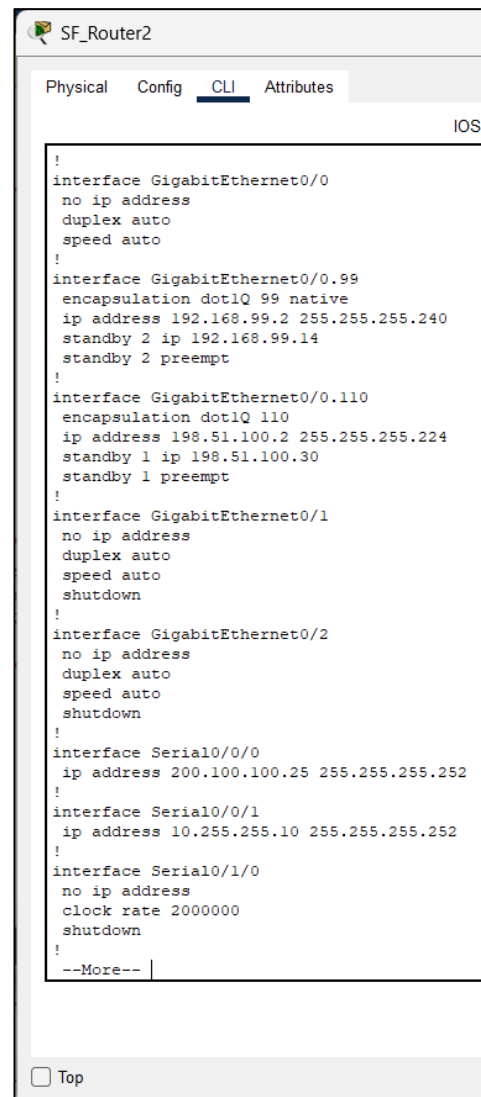
The screenshot shows the CLI configuration for SF_Router1. The tabs at the top are Physical, Config, CLI (selected), and Attributes. The IOS prompt is visible. The configuration includes several interfaces: GigabitEthernet0/0 (no IP), GigabitEthernet0/0.99 (VLAN 99 native, IP 192.168.99.1, standby 2), GigabitEthernet0/0.110 (VLAN 110, IP 198.51.100.1, standby 1), GigabitEthernet0/1 (shutdown), GigabitEthernet0/2 (shutdown), Serial0/0/0 (IP 200.100.100.21), Serial0/0/1 (IP 10.255.255.9, clock rate 64000), and Serial0/1/0 (shutdown). A --More-- prompt is at the bottom.

```

!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.99
encapsulation dot1Q 99 native
ip address 192.168.99.1 255.255.255.240
standby 2 ip 192.168.99.14
standby 2 priority 110
standby 2 preempt
!
interface GigabitEthernet0/0.110
encapsulation dot1Q 110
ip address 198.51.100.1 255.255.255.224
standby 1 ip 198.51.100.30
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/0/0
ip address 200.100.100.21 255.255.255.252
!
interface Serial0/0/1
ip address 10.255.255.9 255.255.255.252
clock rate 64000
!
interface Serial0/1/0
--More--

```

Figure 38 SF Router 1



The screenshot shows the CLI configuration for SF_Router2. The tabs at the top are Physical, Config, CLI (selected), and Attributes. The IOS prompt is visible. The configuration includes several interfaces: GigabitEthernet0/0 (no IP), GigabitEthernet0/0.99 (VLAN 99 native, IP 192.168.99.2, standby 2), GigabitEthernet0/0.110 (VLAN 110, IP 198.51.100.2, standby 1), GigabitEthernet0/1 (shutdown), GigabitEthernet0/2 (shutdown), Serial0/0/0 (IP 200.100.100.25), Serial0/0/1 (IP 10.255.255.10), Serial0/1/0 (shutdown), and a --More-- prompt at the bottom.

```

!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.99
encapsulation dot1Q 99 native
ip address 192.168.99.2 255.255.255.240
standby 2 ip 192.168.99.14
standby 2 preempt
!
interface GigabitEthernet0/0.110
encapsulation dot1Q 110
ip address 198.51.100.2 255.255.255.224
standby 1 ip 198.51.100.30
standby 1 preempt
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/0/0
ip address 200.100.100.25 255.255.255.252
!
interface Serial0/0/1
ip address 10.255.255.10 255.255.255.252
!
interface Serial0/1/0
no ip address
clock rate 2000000
shutdown
!
--More--

```

Figure 37 SF Router 2

In Johor Server Farm, there are two router which is an active router (SF_Router1) and a standby router (SF_Router2). As the figure shows that, the sub interface G0/0/0.99 is encapsulated for native VLAN with dot1Q native. Moreover, the sub interface G0/0/0.110 is configured for Johor server farm.

4.7 HSRP (Hot Standby Router Protocol)

In Cloud Co, HSRP (Hot Standby Router Protocol) has implemented in KL, Penang, Johor, and Johor server farm network. Each branch has configured two router which are an active router and a standby router. This keeps Cloud Co network running even if the active router downs, by automatically switching to standby router. It makes network continuity with no manual intervention. HSRP is implemented in every single VLAN except for Blackhole VLAN and Native VLAN of Penang branch.

4.7.1 KL

```
KL_Router1#sh standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State    Active        Standby        Virtual IP
Gig         1    110 P Active    local         192.168.10.2   192.168.10.62
Gig         2    110 P Active    local         192.168.10.66 192.168.10.126
Gig         3    110 P Active    local         192.168.10.130 192.168.10.158
Gig         4    110 P Active    local         192.168.10.162 192.168.10.174
KL_Router1#
```

Figure 39 KL Router 1 <show standby brief>

```
KL_Router2#sh standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State    Active        Standby        Virtual IP
Gig         1    100 P Standby   192.168.10.1   local         192.168.10.62
Gig         2    100 P Standby   192.168.10.65  local         192.168.10.126
Gig         3    100 P Standby   192.168.10.129 local         192.168.10.158
Gig         4    100 P Standby   192.168.10.161 local         192.168.10.174
KL_Router2#
```

Figure 40 KL Router 2 <show standby brief>

In KL network, the active router (KL_Router1) is configured with a higher priority (110). At the same time, the standby router (KL_Router2) is configured with a lower priority (100). They are all configured in interface GigabitEthernet.

The table below shows the virtual IP, active IP, and standby IP for each department:

	VLAN ID	HSRP Virtual IP	HSRP Active IP	HSRP Standby IP
KL_Management	10	192.168.10.62	192.168.10.1	192.168.10.2
KL_HumanResource	20	192.168.10.126	192.168.10.65	192.168.10.66
KL_Design	30	192.168.10.158	192.168.10.129	192.168.10.130
Native VLAN	99	192.168.10.174	192.168.10.161	192.168.10.162

4.7.2 Penang

```
PG_Router1#sh standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State    Active        Standby        Virtual IP
Gig         1    110 P Active    local         172.16.10.3    172.16.10.1
Gig         2    110 P Active    local         172.16.10.67   172.16.10.65
Gig         3    110 P Active    local         172.16.10.131  172.16.10.129
PG_Router1#
```

Figure 41 PG Router 1 <show standby brief>

```
PG_Router2#sh standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State    Active        Standby        Virtual IP
Gig         1    100 P Standby   172.16.10.2    local         172.16.10.1
Gig         2    100 P Standby   172.16.10.66   local         172.16.10.65
Gig         3    100 P Standby   172.16.10.130  local         172.16.10.129
PG_Router2#
```

Figure 42 PG Router 2 <show standby brief>

In Penang network, the active router (PG_Router1) is configured with a higher priority (110). At the same time, the standby router (PG_Router2) is configured with a lower priority (100). They are all configured interface GigabitEthernet.

The table below shows the virtual IP, active IP, and standby IP for each department:

	VLAN ID	HSRP Virtual IP	HSRP Active IP	HSRP Standby IP
Delivery	60	172.16.10.1	172.16.10.2	172.16.10.3
Management	70	172.16.10.65	172.16.10.66	172.16.10.67
IT	80	172.16.10.129	172.16.10.130	172.16.10.131

4.7.3 Johor

```
JB_Router1#sh standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State    Active        Standby        Virtual IP
Fa          1    110 P Active    local         10.10.1.2      10.10.1.62
Fa          2    110 P Active    local         10.10.1.66     10.10.1.126
Fa          3    110 P Active    local         10.10.1.130    10.10.1.142
Fa          4    110 P Active    local         10.10.1.146    10.10.1.158
JB_Router1#
```

Figure 43 JB Router 1 <show standby brief>

```
JB_Router2#sh standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State    Active        Standby        Virtual IP
Gig         1    100 P Standby   10.10.1.1     local         10.10.1.62
Gig         2    100 P Standby   10.10.1.65    local         10.10.1.126
Gig         3    100 P Standby   10.10.1.129   local         10.10.1.142
Gig         4    100 P Standby   10.10.1.145   local         10.10.1.158
JB_Router2#
```

Figure 44 JB Router 2 <show standby brief>

In Johor network, the active router (JB_Router1) is configured with a higher priority (110) in interface FastEthernet. At the same time, the standby router (JB_Router2) is configured with a lower priority (100) in interface GigabitEthernet.

The table below shows the virtual IP, active IP, and standby IP for each department:

	VLAN ID	HSRP Virtual IP	HSRP Active IP	HSRP Standby IP
Finance	40	10.10.1.62	10.10.1.1	10.10.1.2
R&D	50	10.10.1.126	10.10.1.65	10.10.1.66
R&D_Voice VLAN	60	10.10.1.142	10.10.1.129	10.10.1.130
Native VLAN	99	10.10.1.158	10.10.1.145	10.10.1.146

4.7.4 Server Farm

```
SF_Router1#sh standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State    Active        Standby        Virtual IP
Gig         2    110 P Active    local         192.168.99.2   192.168.99.14
Gig         1    110 P Active    local         198.51.100.2   198.51.100.30
SF_Router1#
```

Figure 45 SF Router 1 <show standby brief>

```
SF_Router2#sh standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State    Active        Standby        Virtual IP
Gig         2    100 P Standby   192.168.99.1   local         192.168.99.14
Gig         1    100 P Standby   198.51.100.1   local         198.51.100.30
SF_Router2#
```

Figure 46 SF Router 2 <show standby brief>

In Johor Server Farm network, the active router (SF_Router1) is configured with a higher priority (110). At the same time, the standby router (SF_Router2) is configured with a lower priority (100). They are all configured interface GigabitEthernet.

The table below shows the virtual IP, active IP, and standby IP for each department:

	VLAN ID	HSRP Virtual IP	HSRP Active IP	HSRP Standby IP
FTP, DNS, Web Server	110	198.51.100.30	198.51.100.1	198.51.100.2
Native VLAN	99	192.168.99.14	192.168.99.1	192.168.99.2

5.0 WAN

5.1 OSPF (Open Shortest Path First)

Open Shortest Path First (OSPF) is a link-state dynamic routing protocol that calculates the shortest path using Dijkstra's algorithm. In our topology OSPF was used to interconnect all routers across Kuala Lumpur (HQ), Johor, Penang, the Server Farm and ISP Routers. The design Area 0 which is the backbone area was used to simply operations. The purpose of configuring OSPF is to automate route change between all branch routers and reduce manual configurations compared to static routing. The <show ip protocol> command is used to verify the OSPF configurations.

The following commands were used to configuring OSPF for every router with different Ip addresses, wildcard mask and passive interfaces.

```
KL_Router1>en
Password:
KL_Router1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
KL_Router1(config)#router ospf 1
KL_Router1(config-router)#default-information originate
KL_Router1(config-router)#router-id 1.1.1.1
KL_Router1(config-router)#network 192.168.10.0 0.0.0.63 area 0
KL_Router1(config-router)#network 192.168.10.64 0.0.0.63 area 0
KL_Router1(config-router)#network 192.168.10.128 0.0.0.31 area 0
KL_Router1(config-router)#network 192.168.10.160 0.0.0.15 area 0
KL_Router1(config-router)#network 200.100.100.0 0.0.0.3 area 0
KL_Router1(config-router)#network 10.255.255.0 0.0.0.3 area 0
KL_Router1(config-router)#network 10.255.255.0 0.0.0.3 area 0
KL_Router1(config-router)#passive-interface g0/0.10
KL_Router1(config-router)#passive-interface g0/0.20
KL_Router1(config-router)#passive-interface g0/0.30
KL_Router1(config-router)#passive-interface g0/0.99
KL_Router1(config-router)#end
```

Figure 47 OSPF Configuration Commands

The command <en> enables privileged EXEC mode to allow configuration changes, while <conf t> enters global configuration mode to allow editing of system settings. The command router <ospf 1> activates the OSPF routing process using process ID 1. The <default-information originate> command is used to advertise a default route into OSPF, allowing other routers to use this device as a gateway to external networks. This command is only used for the active routers. Next, <router-id 1.1.1.1> sets a unique identifier for the router within the OSPF domain, aiding in neighbor formation and routing table management.

The series of <network> commands specify which subnets and interfaces will participate in OSPF, each using a wildcard mask to define the range of IP addresses and placing them in Area 0, the backbone area. To enhance efficiency and security, the passive-interface command is applied on sub interfaces like g0/0.10 and g0/0.20 ensuring their networks are advertised without sending Hello packets. Finally, the <end> command exits configuration mode and returns the router to privileged EXEC mode.

5.1.1 KL

```
KL_Router1>show ip protocol

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  It is an autonomous system boundary router
  Redistributing External Routes from,
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.10.0 0.0.0.63 area 0
    192.168.10.64 0.0.0.63 area 0
    192.168.10.128 0.0.0.31 area 0
    192.168.10.160 0.0.0.15 area 0
    200.100.100.0 0.0.0.3 area 0
    10.255.255.0 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0.10
    GigabitEthernet0/0.20
    GigabitEthernet0/0.30
    GigabitEthernet0/0.99
```

Figure 48 KL_Router1 <show ip protocol>

The <show ip protocol> command on KL_Router1 shows that OSPF process 1 is running with Router ID 1.1.1.1. The router is advertising the networks 192.168.10.0/26, 192.168.10.64/26, 192.168.10.128/27, 192.168.160/28, 200.100.100.0/30 and 10.255.255.0/30 into Area 0. Passive interfaces are G0/0.10, G0/0.20, G0/0.30 and G0/0.99.

```
KL_Router2>show ip protocol

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.10.0 0.0.0.63 area 0
    192.168.10.64 0.0.0.63 area 0
    192.168.10.128 0.0.0.31 area 0
    192.168.10.160 0.0.0.15 area 0
    200.100.100.4 0.0.0.3 area 0
    10.255.255.0 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0.10
    GigabitEthernet0/0.20
    GigabitEthernet0/0.30
    GigabitEthernet0/0.99
```

Figure 49 KL_Router2 <show ip protocol>

The <show ip protocol> command on KL_Router2 shows that OSPF process 1 is running with Router ID 1.1.1.2. The router is advertising the networks 192.168.10.0/26, 192.168.10.64/26, 192.168.10.128/27, 192.168.160/28, 200.100.100.4/30 and 10.255.255.0/30 into Area 0. Passive interfaces are G0/0.10, G0/0.20, /0.30 and G0/0.99.

5.1.2 Penang

```
PG_Router1>show ip protocol

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 2.2.2.1
  It is an autonomous system boundary router
  Redistributing External Routes from,
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    200.100.100.8 0.0.0.3 area 0
    10.255.255.12 0.0.0.3 area 0
    172.16.10.0 0.0.0.63 area 0
    172.16.10.64 0.0.0.63 area 0
    172.16.10.128 0.0.0.31 area 0
  Passive Interface(s):
    GigabitEthernet0/0.60
    GigabitEthernet0/0.80
```

Figure 50 PG_Router1 <show ip protocol>

The <show ip protocol> command on PG_Router1 shows that OSPF process 1 is running with Router ID 2.2.2.1. The router is advertising the networks 200.100.100.8/30, 10.255.255.12/30, 172.16.10.0/26, 172.16.10.64/26 and 172.16.10.128/27 into Area 0. Passive interfaces are G0/0.60 and G0/0.80.

```
PG_Router2>show ip protocol

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 2.2.2.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    200.100.100.12 0.0.0.3 area 0
    10.255.255.12 0.0.0.3 area 0
    172.16.10.0 0.0.0.63 area 0
    172.16.10.64 0.0.0.63 area 0
    172.16.10.128 0.0.0.31 area 0
  Passive Interface(s):
    GigabitEthernet0/0.60
    GigabitEthernet0/0.80
```

Figure 51 PG_Router2 <show ip protocol>

The <show ip protocol> command on PG_Router2 shows that OSPF process 1 is running with Router ID 2.2.2.2. The router is advertising the networks 200.100.100.12/30, 10.255.255.12/30, 172.16.10.0/26, 172.16.10.64/26 and 172.16.10.128/27 into Area 0. Passive interfaces are G0/0.60 and G0/0.80.

5.1.3 Johor

```
JB_Router1>show ip protocol

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.3.3.1
  It is an autonomous system boundary router
  Redistributing External Routes from,
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.1.0 0.0.0.63 area 0
    10.10.1.64 0.0.0.63 area 0
    10.10.1.128 0.0.0.15 area 0
    10.10.1.144 0.0.0.15 area 0
    200.100.100.28 0.0.0.3 area 0
    10.255.255.4 0.0.0.3 area 0
  Passive Interface(s):
    FastEthernet0/0.40
    FastEthernet0/0.50
    FastEthernet0/0.60
    FastEthernet0/0.99
```

Figure 52 JB_Router1 <show ip protocol>

The <show ip protocol> command on JB_Router1 shows that OSPF process 1 is running with Router ID 3.3.3.1. The router is advertising the networks 10.10.1.0/26, 10.10.1.64/26, 10.10.1.128/30, 10.10.1.144/28, 200.100.100.28/30 and 10.255.255.4/30 into Area 0. Passive interfaces are G0/0.40, G0/0.50, G0/0.60 and G0/0.99.

```
JB_Router2>show ip protocol

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.3.3.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.1.0 0.0.0.63 area 0
    10.10.1.64 0.0.0.63 area 0
    10.10.1.128 0.0.0.15 area 0
    10.10.1.144 0.0.0.15 area 0
    200.100.100.32 0.0.0.3 area 0
    10.255.255.4 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0.40
    GigabitEthernet0/0.50
    GigabitEthernet0/0.60
    GigabitEthernet0/0.99
```

Figure 53 JB_Router2 <show ip protocol>

The <show ip protocol> command on JB_Route2 shows that OSPF process 1 is running with Router ID 3.3.3.2. The router is advertising the networks 10.10.1.0/26, 10.10.1.64/26, 10.10.1.128/30, 10.10.1.144/28, 200.100.100.32/30 and 10.255.255.4/30 into Area 0. Passive interfaces are G0/0.40, G0/0.50, G0/0.60 and G0/0.99.

5.1.4 Server Farm

```
SF_Router1>show ip protocol

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 4.4.4.1
  It is an autonomous system boundary router
  Redistributing External Routes from,
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    198.51.100.0 0.0.0.31 area 0
    192.168.99.0 0.0.0.15 area 0
    200.100.100.20 0.0.0.3 area 0
    10.255.255.8 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0.99
    GigabitEthernet0/0.110
```

Figure 54 SF_Router1 <show ip protocol>

The <show ip protocol> command on SF_Route1 shows that OSPF process 1 is running with Router ID 4.4.4.1. The router is advertising the networks 198.51.100.0/27, 192.168.99.0/28, 200.100.100.20/30 and 10.255.255.8/30 into Area 0. Passive interfaces are G0/0.99 and G 0/0.110.

```
SF_Router2>show ip pro

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 4.4.4.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    198.51.100.0 0.0.0.31 area 0
    192.168.99.0 0.0.0.15 area 0
    200.100.100.24 0.0.0.3 area 0
    10.255.255.8 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0.99
    GigabitEthernet0/0.110
```

Figure 55 SF_Router2 <show ip protocol>

The <show ip protocol> command on SF_Route2 shows that OSPF process 1 is running with Router ID 4.4.4.2. The router is advertising the networks 198.51.100.0/27, 192.168.99.0/28, 200.100.100.24/30 and 10.255.255.8/30 into Area 0. Passive interfaces are G0/0.99 and G 0/0.110.

5.1.5 ISP

```
ISP1>show ip protocol

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 5.5.5.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    200.100.100.0 0.0.0.3 area 0
    200.100.100.8 0.0.0.3 area 0
    200.100.100.16 0.0.0.3 area 0
    200.100.100.24 0.0.0.3 area 0
    200.100.100.32 0.0.0.3 area 0
```

Figure 56 ISP1 <show ip protocol>

The <show ip protocol> command on ISP1 shows that OSPF process 1 is running with Router ID 5.5.5.1. The router is advertising the networks 200.100.100.0/30, 200.100.100.8/30, 200.100.100.16/30, 200.100.100.24/30 and 200.100.100.32/30 into Area 0. All the connected links are fully participating in OSPF, allowing ISP1 to exchange routes with other routers across the WAN.

```
ISP2>show ip protocol

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 5.5.5.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    200.100.100.4 0.0.0.3 area 0
    200.100.100.12 0.0.0.3 area 0
    200.100.100.16 0.0.0.3 area 0
    200.100.100.20 0.0.0.3 area 0
    200.100.100.28 0.0.0.3 area 0
```

Figure 57 ISP2 <show ip protocol>

The <show ip protocol> command on ISP2 shows that OSPF process 1 is running with Router ID 5.5.5.2. The router is advertising the networks 200.100.100.4/30, 200.100.100.12/30, 200.100.100.16/30, 200.100.100.20/30 and 200.100.100.28/30 into Area 0. All the connected links are fully participating in OSPF, allowing ISP2 to exchange routes with other routers across the WAN.

5.2 Default Static Route and Floating Static Route

While OSPF handles internal dynamic routing, static routes are configured to manage external traffic and redundancy. A default route forwards unknown traffic to the ISP, while a floating static route provides backup path in case of link failure.

The following commands were used to configure default and floating static route for every router with different Ip addresses.

```
KL_Router1>en
Password:
KL_Router1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
KL_Router1(config)#ip route 0.0.0.0 0.0.0.0 200.100.100.2
KL_Router1(config)#ip route 0.0.0.0 0.0.0.0 10.255.255.2 200
```

Figure 58: Default and Floating Static Route Configurations

The command <en> enables privileged EXEC mode to allow configuration changes, while <conf t> enters global configuration mode to allow editing of system settings. The command <ip route 0.0.0.0 0.0.0.0 200.100.100.2> creates a default static route that forwards all traffic with unknown destinations to the next-hop address 200.100.100.2. This ensures that external traffic can exit the local network. The command <0.0.0.0 0.0.0.0 10.255.255.2 200> configures a floating using next hop 10.255.255.255.2 with an administrative distance of 200 making it a backup path. It will only be used if the primary default route becomes unavailable. Thus providing redundancy.

5.2.1 KL

```
KL_Router1#show running-config | include ip route
ip route 0.0.0.0 0.0.0.0 200.100.100.2
ip route 0.0.0.0 0.0.0.0 10.255.255.2 200
```

Figure 59 KL_Router1 <show running-config | include ip route>

The output from the command <show running-config | include ip route> on KL_Router1 shows ip route 0.0.0.0 0.0.0.0 200.100.100.2 as the default static route and ip route 0.0.0.0 0.0.0.0 10.255.255.2 200 as the floating route with administrative distance of 200.

```
KL_Router2#show running-config | include ip route
ip route 0.0.0.0 0.0.0.0 200.100.100.6
ip route 0.0.0.0 0.0.0.0 10.255.255.1 200
```

Figure 60 KL_Router2 <show running-config | include ip route>

The output from the command <show running-config | include ip route> on KL_Router2 shows ip route 0.0.0.0 0.0.0.0 200.100.100.6 as the default static route and ip route 0.0.0.0 0.0.0.0 10.255.255.1 200 as the floating route with administrative distance of 200.

5.2.2 Penang

```
PG_Router1#show running-config | include ip route
ip route 0.0.0.0 0.0.0.0 200.100.100.10
ip route 0.0.0.0 0.0.0.0 10.255.255.14 200
```

Figure 61 PG_Router1 <show running-config | include ip route>

The output from the command <show running-config | include ip route> on PG_Router1 shows ip route 0.0.0.0 0.0.0.0 200.100.100.10 as the default static route and ip route 0.0.0.0 0.0.0.0 10.255.255.14 200 as the floating route with administrative distance of 200.

```
PG_Router2#show running-config | include ip route
ip route 0.0.0.0 0.0.0.0 200.100.100.14
ip route 0.0.0.0 0.0.0.0 10.255.255.13 200
```

Figure 62 PG_Router2 <show running-config | include ip route>

The output from the command <show running-config | include ip route> on PG_Router2 shows ip route 0.0.0.0 0.0.0.0 200.100.100.14 as the default static route and ip route 0.0.0.0 0.0.0.0 10.255.255.13 200 as the floating route with administrative distance of 200.

5.2.3 Johor

```
JB_Router1#show running-config | include ip route  
ip route 0.0.0.0 0.0.0.0 200.100.100.30  
ip route 0.0.0.0 0.0.0.0 10.255.255.6 200
```

Figure 63 JB_Router1 <show running-config | include ip route>

The output from the command <show running-config | include ip route> on JB_Router1 shows ip route 0.0.0.0 0.0.0.0 200.100.100.30 as the default static route and ip route 0.0.0.0 0.0.0.0 10.255.255.6 200 as the floating route with administrative distance of 200.

```
JB_Router2#show running-config | include ip route  
ip route 0.0.0.0 0.0.0.0 200.100.100.34  
ip route 0.0.0.0 0.0.0.0 10.255.255.5 200
```

Figure 64 JB_Router2 <show running-config | include ip route>

The output from the command <show running-config | include ip route> on JB_Router2 shows ip route 0.0.0.0 0.0.0.0 200.100.100.34 as the default static route and ip route 0.0.0.0 0.0.0.0 10.255.255.5 200 as the floating route with administrative distance of 200.

5.2.4 Server Farm

```
SF_Router1#show running-config | include ip route
ip route 0.0.0.0 0.0.0.0 200.100.100.22
ip route 0.0.0.0 0.0.0.0 10.255.255.10 200
```

Figure 65 SF_Router1 <show running-config | include ip route>

The output from the command <show running-config | include ip route> on SG_Router1 shows ip route 0.0.0.0 0.0.0.0 200.100.100.22 as the default static route and ip route 0.0.0.0 0.0.0.0 10.255.255.10 200 as the floating route with administrative distance of 200.

```
SF_Router2#show running-config | include ip route
ip route 0.0.0.0 0.0.0.0 200.100.100.26
ip route 0.0.0.0 0.0.0.0 10.255.255.9 200
```

Figure 66 SF_Router2 <show running-config | include ip route>

The output from the command <show running-config | include ip route> on SF_Router2 shows ip route 0.0.0.0 0.0.0.0 200.100.100.26 as the default static route and ip route 0.0.0.0 0.0.0.0 10.255.255.9 200 as the floating route with administrative distance of 200.

6.0 Penang WLAN

6.1 Overall Architecture Description

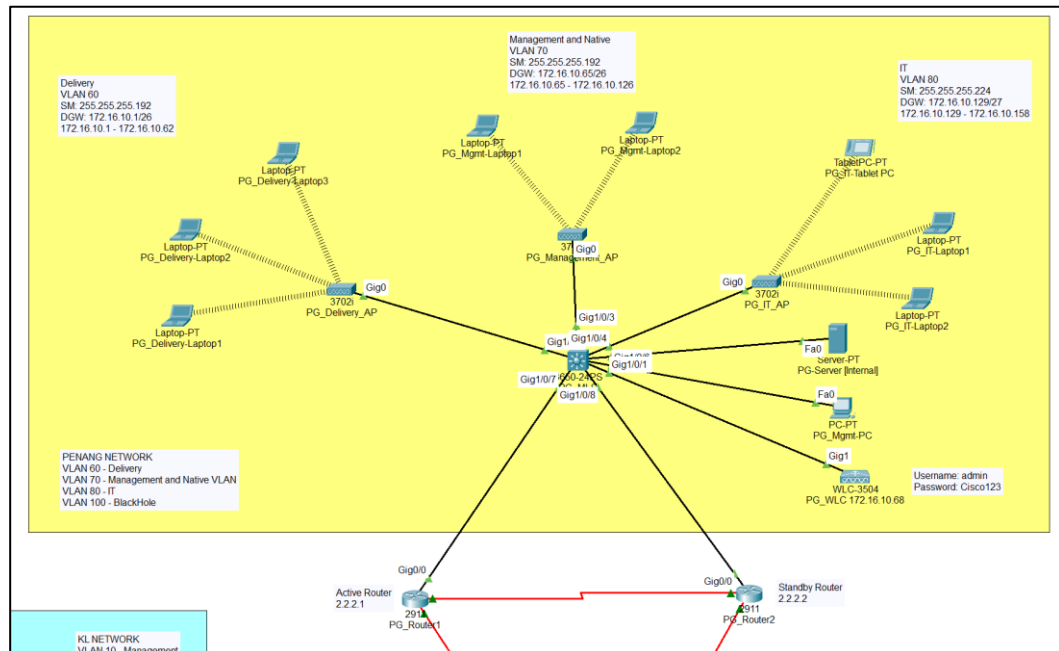


Figure 67 Penang WLAN

A WLAN has been installed in the Penang network to support secure wireless connectivity among various departments. The central management of the WLAN is a Cisco 3504 WLC, and the access points (APs) are distributed in areas of Delivery, Management, and IT. Every WLAN is projected to a VLAN, so that the users are assigned appropriate IP addresses and are not taken out of the network segment they belong to.

DHCP and IP Assignment

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
PG_IT	172.16.10.129	198.51.100.6	172.16.10.139	255.255.255.224	21	0.0.0.0	172.16.10.68
PG_Mgmt	172.16.10.65	198.51.100.6	172.16.10.75	255.255.255.192	50	0.0.0.0	172.16.10.68
PG_Delivery	172.16.10.1	198.51.100.6	172.16.10.10	255.255.255.192	54	0.0.0.0	172.16.10.68

Figure 68 DHCP and IP Assignment

The Johor Server Farm offers DHCP services, and the routers apply IP helper-address in relaying the requests and allocating IPs to wireless end devices.

6.2 WLC Configuration

Interfaces

Interfaces					
Interface Name	VLAN Identifier	IP Address	Interface Type	Dynamic AP Management	IPv6 Address
Delivery	60	172.16.10.5	Dynamic	Disabled	
IT	80	172.16.10.133	Dynamic	Disabled	
management	1	172.16.10.68	Static	Enabled	::/128
virtual	N/A	192.0.2.1	Static	Not Supported	

Figure 69 Interfaces Name

Interfaces > Edit

General Information

Interface Name

Delivery

MAC Address

00:30:F2:4C:13:BB

Configuration

Guest Lan

☐

Quarantine

☐

Quarantine Vlan Id

0

NAS-ID

Physical Information

Port Number

1

Backup Port

0

Active Port

1

Enable Dynamic AP Management

☐

Interface Address

VLAN Identifier

60

IP Address

172.16.10.5

Netmask

255.255.255.192

Gateway

172.16.10.1

DHCP Information

Primary DHCP Server

198.51.100.8

Secondary DHCP Server

DHCP Proxy Mode

Global ▼

Enable DHCP Option 82

☐

Figure 70 Interface Delivery

Interfaces > Edit

General Information

Interface Name: management
 MAC Address: 00:60:70:1E:C8:E3

Configuration

Guest Lan: ☐
 NAS-ID:

NAT Address

Enable NAT Address: ☐

Physical Information

Port Number: 1
 Backup Port: 0
 Active Port: 1
 Enable Dynamic AP Management: ☒

Interface Address

VLAN Identifier: 1
 IP Address: 172.16.10.68
 Netmask: 255.255.255.192
 Gateway: 172.16.10.65

DHCP Information

Primary DHCP Server: 198.51.100.8
 Secondary DHCP Server:
 DHCP Proxy Mode: Global
 Enable DHCP Option 82: ☐

Figure 71 Interface Management

Interfaces > Edit

General Information

Interface Name: IT
 MAC Address: 00:04:9A:45:3A:23

Configuration

Guest Lan: ☐
 Quarantine: ☐
 Quarantine Vlan Id: 0
 NAS-ID:

Physical Information

Port Number: 1
 Backup Port: 0
 Active Port: 1
 Enable Dynamic AP Management: ☐

Interface Address

VLAN Identifier: 80
 IP Address: 172.16.10.133
 Netmask: 255.255.255.224
 Gateway: 172.16.10.129

DHCP Information

Primary DHCP Server: 198.51.100.8
 Secondary DHCP Server:
 DHCP Proxy Mode: Global
 Enable DHCP Option 82: ☐

Figure 72 Interface IT

On the WLC, each interface was configured with specific parameters. For example, Interface Name, VLAN Identifier, IP Address, Netmask, Gateway, and Primary DHCP Server. These interfaces allow WLANs/SSIDs to be mapped to the right VLANs. It ensuring that each wireless network is associated with the intended subnet and applies the corresponding security.

WLANs

WLANs						
Current Filter: [Change Filter] [Clear Filter] Create New Go						Entries 1 - 3 of 3
☐	WLAN ID	Type	Profile Name	WLAN SSID	Admin Status	Security Policies
☐	1	WLAN	PG_Delivery	PG_Delivery	Enabled	[WPA2][Auth(PSK)]
☐	2	WLAN	PG_IT	PG_IT	Enabled	[WPA2][Auth(PSK)]
☐	3	WLAN	PG_Management	PG_Management	Enabled	[WPA2][Auth(PSK)]

Figure 73 WLANs

WLANs > Edit 'PG_Delivery'

General **Security** **QoS** **Policy-Mapping** **Advanced**

Profile Name: PG_Delivery

Type: WLAN

SSID: PG_Delivery

Status: ☒ Enabled

Security Policies: [WPA2][Auth(PSK)]
(Modifications done under security tab will appear after applying the changes.)

Radio Policy: All

Interface/Interface Group(G): Delivery

Multicast Vlan Feature: ☐ Enabled

Broadcast SSID: ☒ Enabled

NAS-ID:

Figure 74 WLANs PG_Delivery General

WLANs > Edit 'PG_Delivery'

General **Security** **QoS** **Policy-Mapping** **Advanced**

Layer 2 **Layer 3** **AAA Servers**

Layer 2 Security: WPA+WPA2

MAC Filtering: ☐

Fast Transition

Fast Transition: ☐

Protected Management Frame

PMF: Disabled

WPA+WPA2 Parameters

WPA Policy: ☐

WPA2 Policy: ☒

WPA2 Encryption: ☒ AES ☐ TKIP

Authentication Key Management

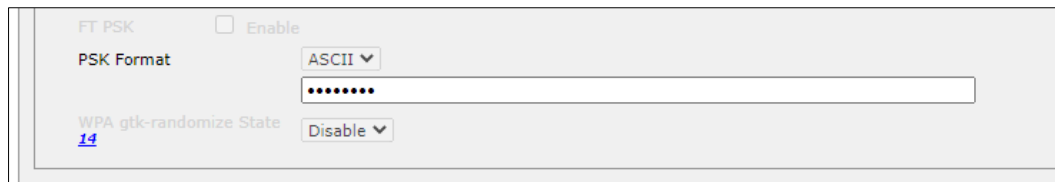
802.1X: ☐ Enable

CCKM: ☐ Enable

PSK: ☒ Enable

FT 802.1X: ☐ Enable

Figure 75 WLANs PG_Delivery Security



The screenshot displays a configuration panel for WLAN settings. At the top, there is a section for 'FT PSK' with an 'Enable' checkbox that is currently unchecked. Below this, the 'PSK Format' is set to 'ASCII' via a dropdown menu. A text input field for the PSK key is shown with ten dots, indicating a masked password. At the bottom of the panel, the 'WPA gtk-randomize State' is set to 'Disable' via another dropdown menu. A small blue link labeled '14' is visible to the left of the 'Disable' dropdown.

Figure 76 WLANs PG_Delivery PSK

Each WLAN was properly configured in the WLAN configuration section of the WLC and its SSID is defined and secured by WPA2-PSK. This setup will provide a high security posture by making sure that only users with the appropriate pre-shared key can use the network and eliminate the chances of unauthorized access to the wireless infrastructure.

AP Groups

AP Groups		Entries 1 - 4 of 4
AP Group Name	AP Group Description	
Delivery_APGroup	PG Delivery Group	Remove
IT_APGroup	PG IT Group	Remove
Management_APGroup	PG Management Group	Remove
default-group		

Figure 77 AP Groups Name

AP Groups > Edit 'Delivery_APGroup'

General
 WLANs
 RF Profile
 APs
 802.11u
 Location
 Ports/Module

APs currently in the Group
 Remove APs

Add APs to the Group
 Add APs

☐ AP Name	Ethernet MAC	☐ AP Name	Group Name
☐ PG_Delivery_AP	000A.F365.8201	☐ PG_IT_AP	IT_APGroup
		☐ PG_Management_AP	Management_APGroup

Figure 78 Delivery_APGroup

Under the AP Groups setup, one department had one groups and assigned APs to their groups. This arrangement will mean that each department only broadcasts its assigned WLAN and will enhance organization and control of access to the network.

6.3 Verification of WLAN Configuration

LAP Registration & CAPWAP

```

Device Name: PG_Delivery_AP
Device Model: 3702i

Port          Link    IP Address      MAC Address
GigabitEthernet0  Up    172.16.10.75/26  000A.F365.8201
Dot11Radio0    Up    <not set>        000A.F365.8202

CAPWAP Status: Connected to 172.16.10.68
Providing WLANs:
    PG_Delivery (PG_Delivery)

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > PG_Delivery_AP

```

Figure 79 PG_Delivery_AP LAP Registration

```

Device Name: PG_Management_AP
Device Model: 3702i

Port          Link    IP Address      MAC Address
GigabitEthernet0  Up    172.16.10.80/26  0000.0CEA.EE01
Dot11Radio0    Up    <not set>        0000.0CEA.EE02

CAPWAP Status: Connected to 172.16.10.68
Providing WLANs:
    PG_Management (PG_Management)

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > PG_Management_AP

```

Figure 80 PG_Management_AP LAP Registration

```

Device Name: PG_IT_AP
Device Model: 3702i

Port          Link    IP Address      MAC Address
GigabitEthernet0  Up    172.16.10.76/26  00E0.8F71.4301
Dot11Radio0    Up    <not set>        00E0.8F71.4302

CAPWAP Status: Connected to 172.16.10.68
Providing WLANs:
    PG_IT (PG_IT)

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > PG_IT_AP

```

Figure 81 PG_IT_AP LAP Registration

Based on CAPWAP, the LAP was able to enroll and create secure control and data tunnels with the WLC. The LAPs were registered with the WLC and attached to the appropriate WLAN by means of AP Groups. An example is that it confirmed by the Providing WLANs: PG_Delivery status. This guaranteed wireless department-wise coverage.

End Device Connectivity Test

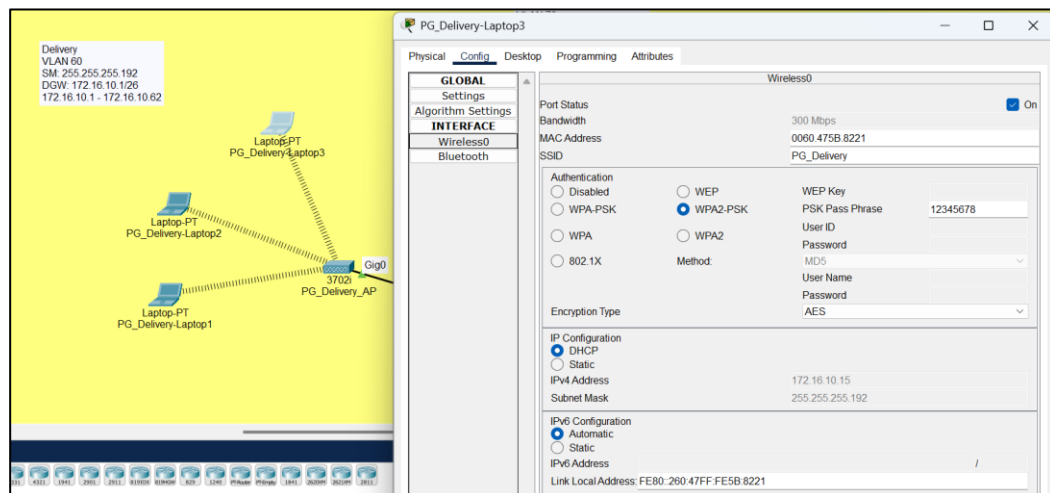


Figure 82 PG_Delivery-Laptop3 Configuration

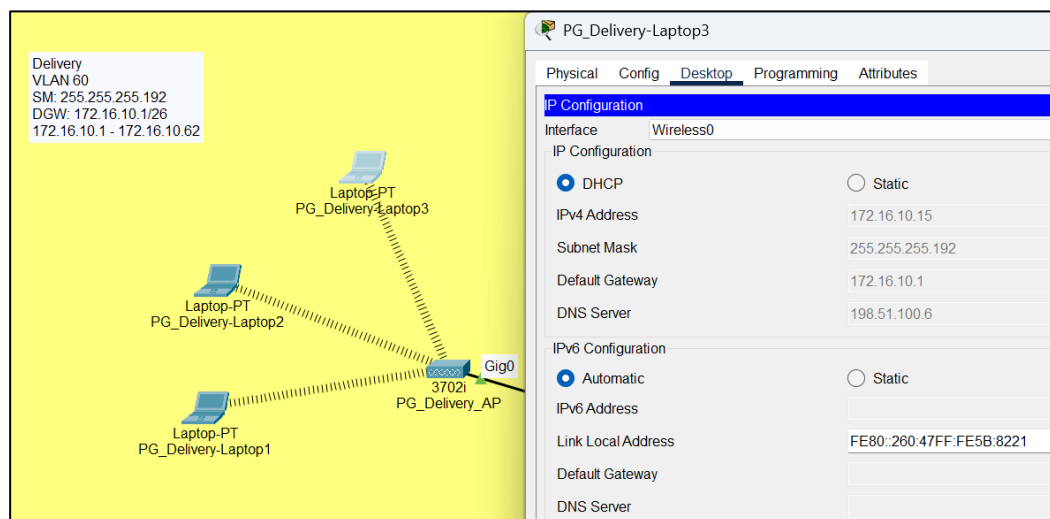


Figure 83 PG_Delivery-Laptop3 Desktop

As part of end device connectivity test, laptops managed to connect to their corresponding LAPs by choosing the appropriate SSID, applying WPA2-PSK authentication and keying in the assigned passphrase. Upon connection the devices automatically got the valid IP addresses provided by the DHCP server. This will in turn fully assure appropriate WLAN functionality and secure access.

7.0 Server Services in Server Farm

7.1 FTP Server

The FTP Server Farm provides the organization with a centralized platform for secure file storage and transfer. It works on a client-server model, where the server stores files and the client uploads or downloads them. It allows users to manage files on a remote system by uploading, downloading, renaming, and deleting files. This service is essential for data exchange, backups, and efficient distribution of files and resources.

7.1.1 Configuration Process

FTP Server has been placed in the Johor Server Farm.

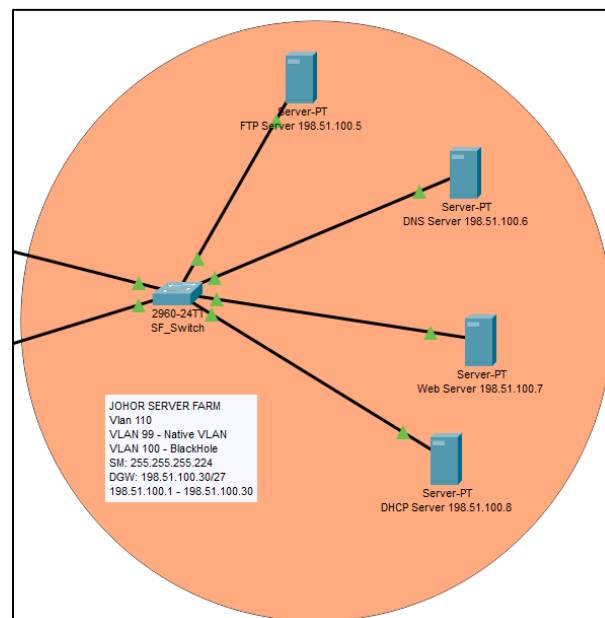
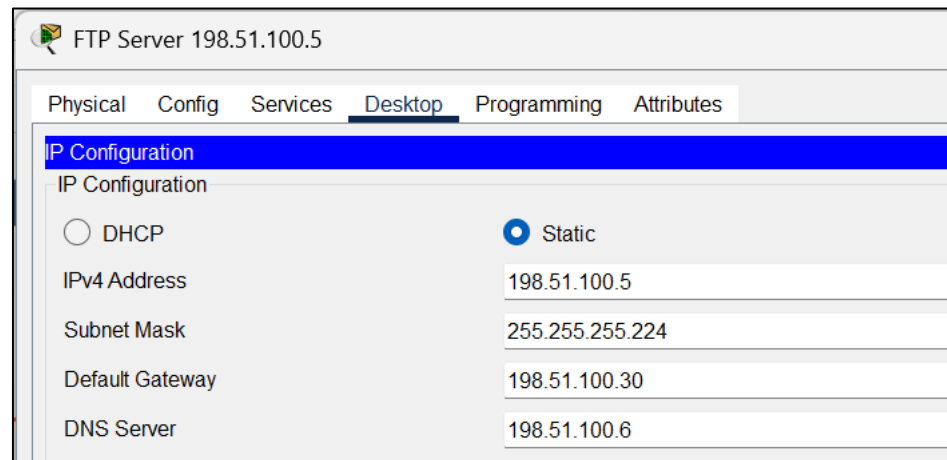


Figure 84 Johor Server Farm

1. Assigned the IPV4 Address (198.51.100.5), subnet mask (255.255.255.224), and default gateway (188.51.100.30) as the vital information in configuring.



FTP Server 198.51.100.5

Physical Config Services **Desktop** Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 198.51.100.5

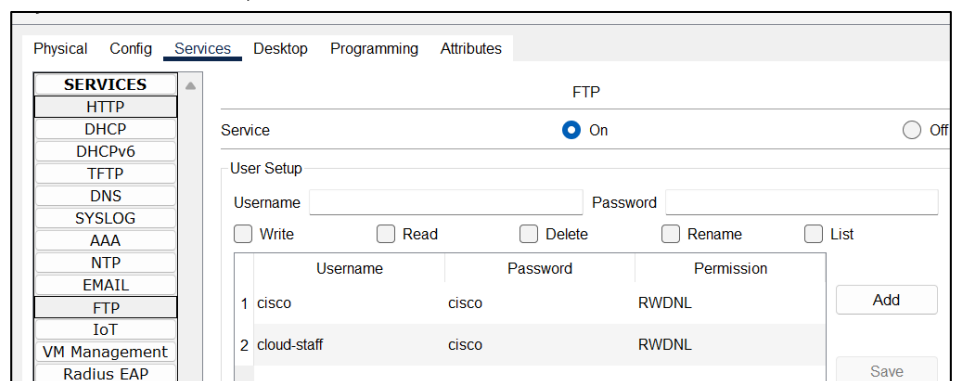
Subnet Mask 255.255.255.224

Default Gateway 198.51.100.30

DNS Server 198.51.100.6

Figure 85 IP Configuration

2. Under the services, ensure that the FTP Service is enabled.



Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP**
- IoT
- VM Management
- Radius EAP

FTP

Service ☒ On ☐ Off

User Setup

Username Password

☐ Write ☐ Read ☐ Delete ☐ Rename ☐ List

	Username	Password	Permission	
1	cisco	cisco	RWDNL	Add
2	cloud-staff	cisco	RWDNL	Save

Figure 86 FTP Service

7.1.2 Verification of FTP Server Configuration

1. Created a new FTP username by allowing all permissions in the FTP Server under the services.

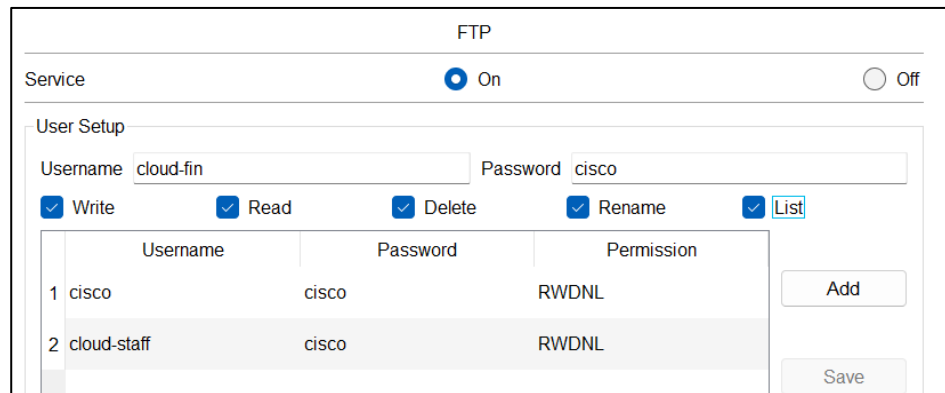


Figure 87 FTP Username

2. Selected a random PC from KL branch and wrote this message into the text editor.

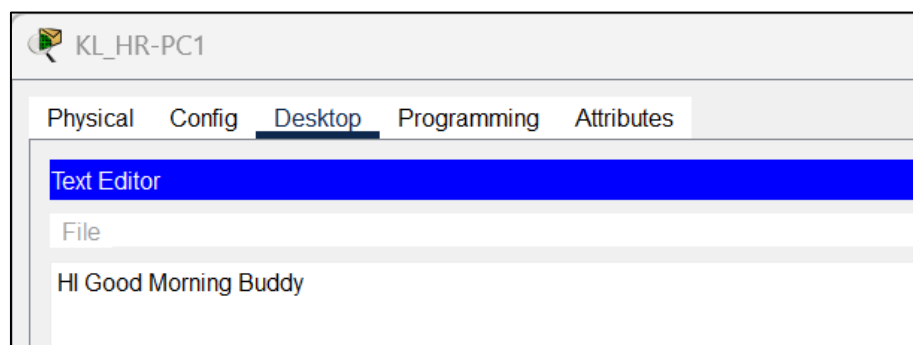


Figure 88 KL_HR-PC1 Message

3. Saved the text editor file as “fin.txt” file.

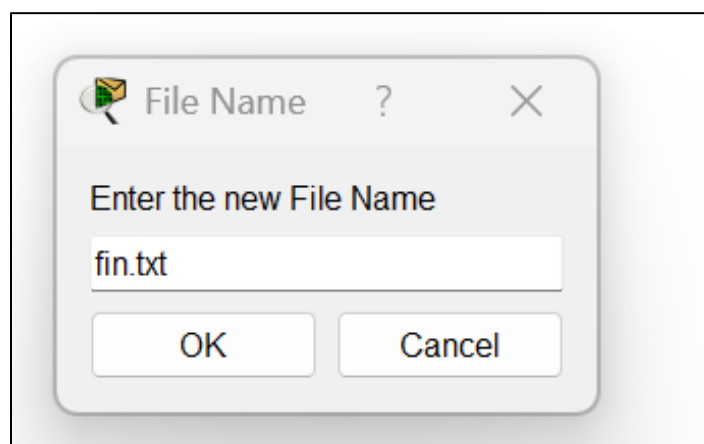
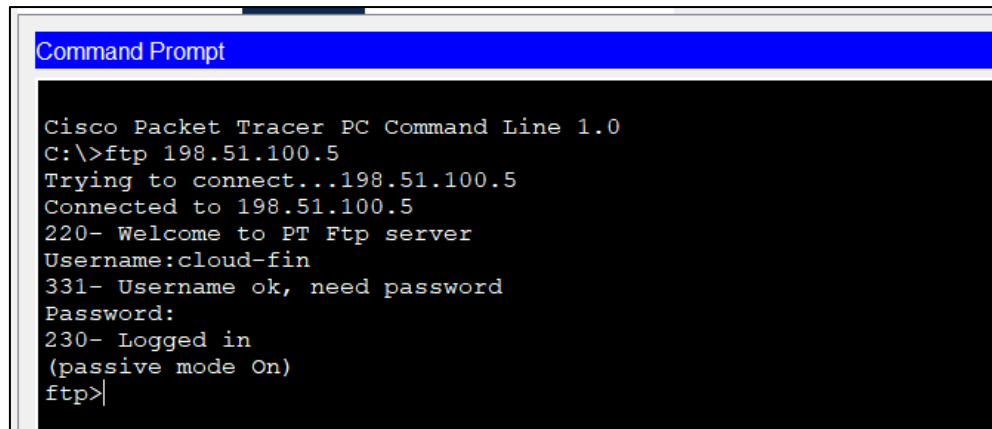


Figure 89 fin.txt file

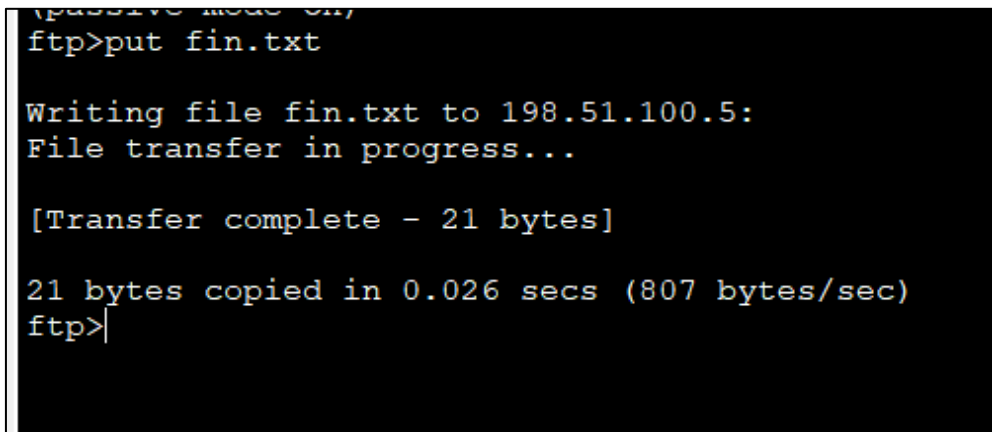
4. In the command prompt, connected to the FTP Server using its IP Address (198.51.100.5) and fill in the username and password of the newly FTP that created.



```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 198.51.100.5
Trying to connect...198.51.100.5
Connected to 198.51.100.5
220- Welcome to PT Ftp server
Username:cloud-fin
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

5. After logging in, use the “put” command to upload the txt file that has been created.



```
(passive mode On)
ftp>put fin.txt

Writing file fin.txt to 198.51.100.5:
File transfer in progress...

[Transfer complete - 21 bytes]

21 bytes copied in 0.026 secs (807 bytes/sec)
ftp>
```

6. Checked the directory to ensure the fin.txt file is in.

```

21 bytes copied in 0.026 secs (807 bytes/sec)
ftp>dir

Listing /ftp directory from 198.51.100.5:
 0 : asa842-k8.bin                    5571584
 1 : asa923-k8.bin                    30468096
 2 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
 3 : c1841-ipbase-mz.123-14.T7.bin    13832032
 4 : c1841-ipbasek9-mz.124-12.bin    16599160
 5 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
 6 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
 7 : c2600-i-mz.122-28.bin            5571584
 8 : c2600-ipbasek9-mz.124-8.bin      13169700
 9 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
10 : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
11 : c2800nm-ipbase-mz.123-14.T7.bin  5571584
12 : c2800nm-ipbasek9-mz.124-8.bin    15522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
14 : c2950-i6q412-mz.121-22.EA4.bin  3058048
15 : c2950-i6q412-mz.121-22.EA8.bin  3117390
16 : c2960-lanbase-mz.122-25.FX.bin   4414921
17 : c2960-lanbase-mz.122-25.SEE1.bin 4670455
18 : c2960-lanbasek9-mz.150-2.SE4.bin 4670455
19 : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
20 : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
21 : c800-universalk9-mz.SPA.152-4.M4.bin 33591768
22 : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
23 : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
24 : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
25 : cgr1000-universalk9-mz.SPA.156-3.CG 184530138
26 : cloud1.txt                      30
27 : fin.txt                          21
28 : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
29 : ir800-universalk9-mz.SPA.155-3.M  61750062
30 : ir800-universalk9-mz.SPA.156-3.M  63753767
31 : ir800_yocto-1.7.2.tar            2877440
32 : ir800_yocto-1.7.2_python-2.7.3.tar 6912000
33 : pt1000-i-mz.122-28.bin            5571584
34 : pt3000-i6q412-mz.121-22.EA4.bin  3117390
ftp>

```

7. Use another PC to get the txt file by using “get” command.

```

(passive mode on)
ftp>get fin.txt

Reading file fin.txt from 198.51.100.5:
File transfer in progress...

[Transfer complete - 21 bytes]

21 bytes copied in 0.031 secs (677 bytes/sec)
ftp>

```

7.2 Web Server

The Web server in Johor Server Farm provides the organization with a centralized platform to host corporate applications, websites, and internal portals. It enables staff and remote branches to access web resources using standard HTTP communication. This service is essential for improving collaboration, delivering intranet services, and providing a foundation for future public-facing applications.

7.2.1 Configuration Process

The Web Server is placed in the Johor Server Farm in Cisco Packet Tracer's server device.

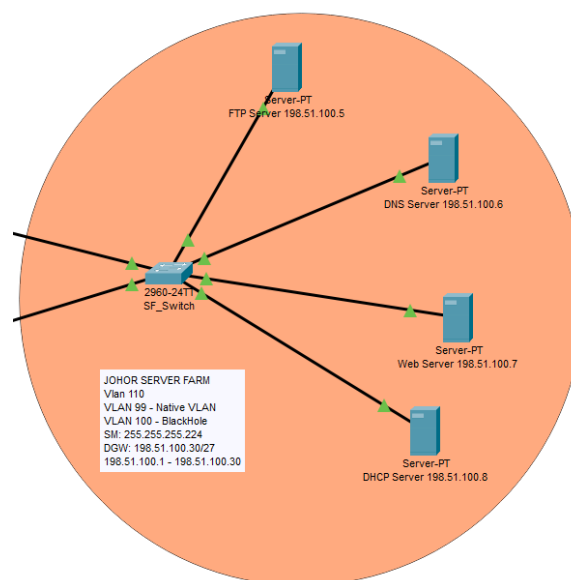


Figure 7.2.1.1 Johor Server Farm

1. Start by assigning a static IPv4 address to the server (198.51.100.7) along with the appropriate subnet mask (255.255.255.224) and gateway (198.51.100.30), in line with the addressing plan.

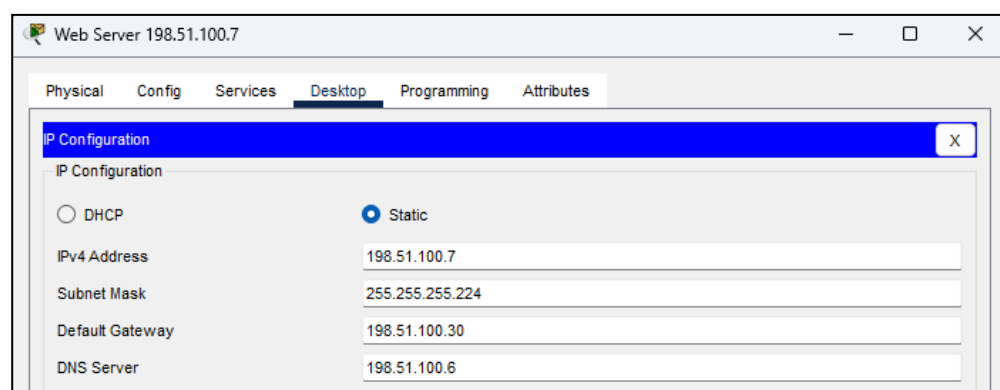


Figure 7.2.1.2 Web Server IP Address

2. Within the server's Services tab, the HTTP service was enabled.

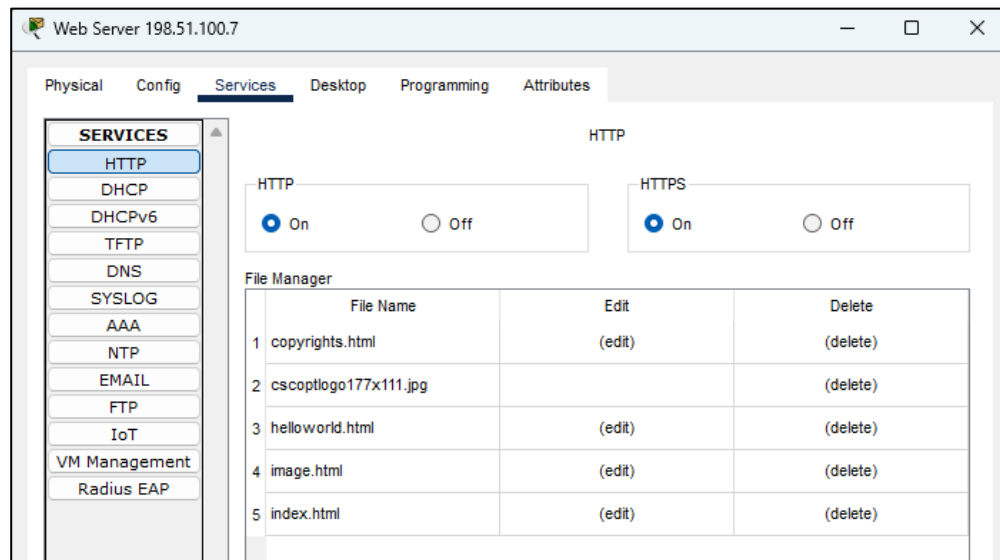


Figure 7.2.1.3 Web Sever HTTP service

3. The default index.html file was edited to include a customized landing page for Cloud Co. For example, we created a webpage using html displaying our group's information.

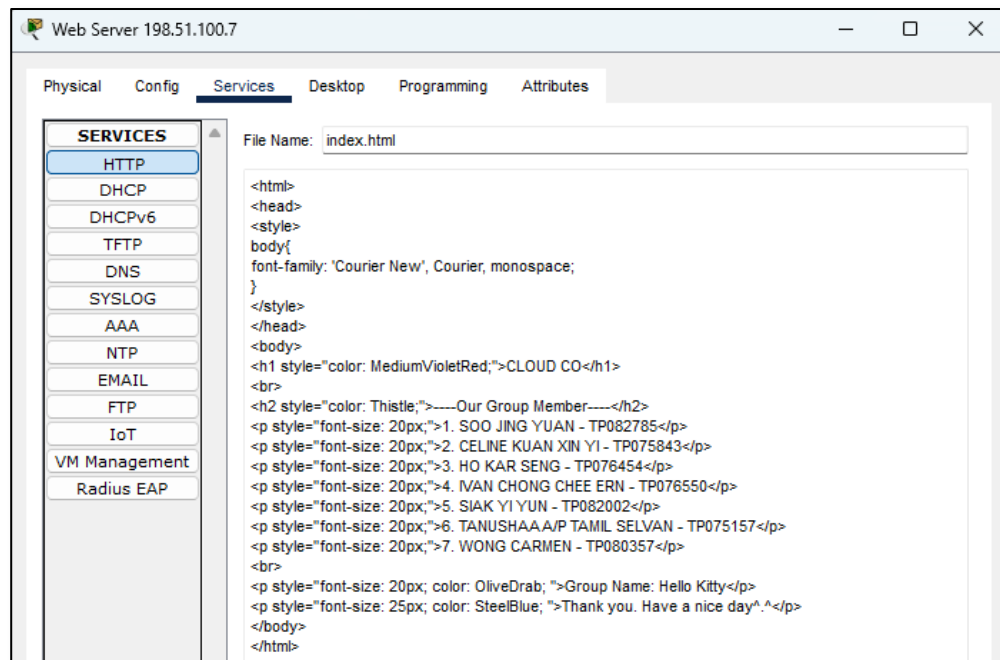


Figure 7.2.1.4 HTML code for a website

4. The configuration was saved, ensuring that the web server remained accessible across sessions.

7.2.2 Verification of Web Server Configuration

To verify the functionality of the configured web server, start by going onto a PC that is connected to the network and enter the Web Browser in the Desktop.

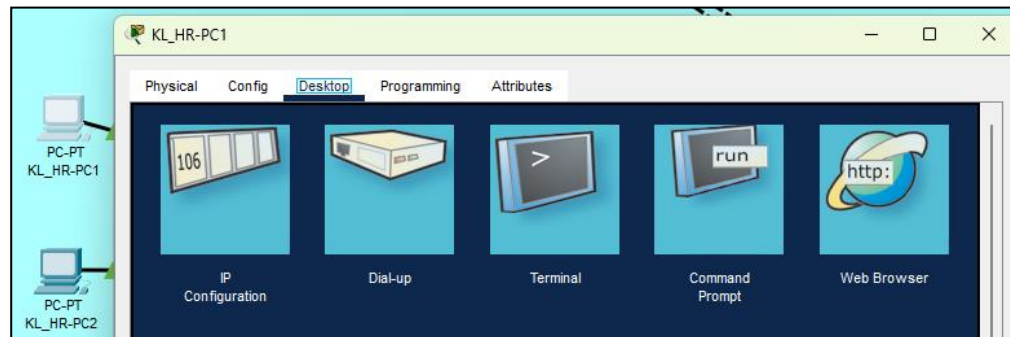


Figure 7.2.2.1 Desktop view of a computer

After opening the Web Browser, enter the IP address of the Web server (198.51.100.7) and click “Go” to try and visit the website configured in the Web server.

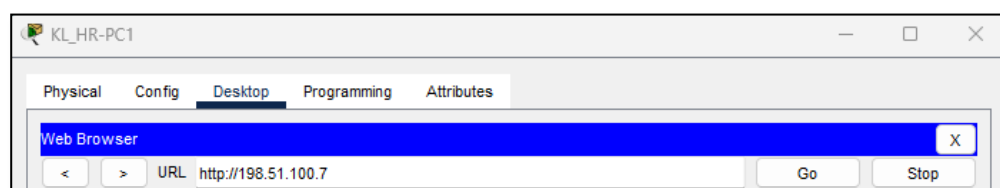


Figure 7.2.2.2 Web Browser view in Desktop

If the website is successfully displayed, this proves the successful configuration of the Web Server.



Figure 7.2.2.3 Web Browser with Web Server IP Address

7.3 DNS Server

The Domain Name System (DNS) server plays an important role in translating user-friendly domain names into corresponding IP addresses. In our Cloud Co network, the DNS server simplifies access to web and FTP services by allowing staff to use meaningful names like (www.cloud.com) instead of numerical IP addresses. This improves usability, reduces errors in accessing resources, and supports scalability for future service deployments.

7.3.1 Configuration Process

The DNS Server is placed in the Johor Server Farm in Cisco Packet Tracer's server device.

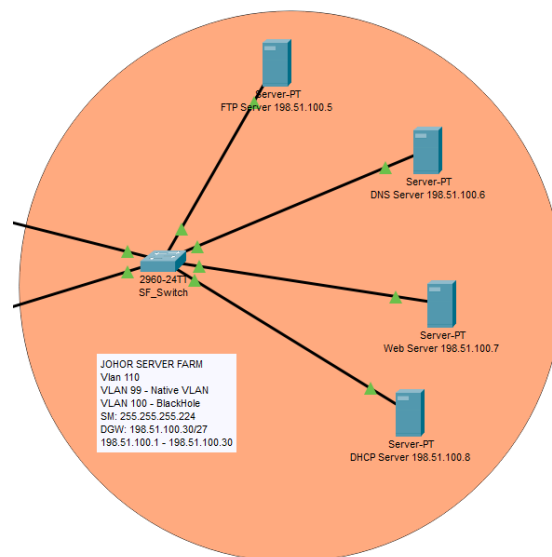


Figure 7.3.1.1 Johor Server Farm

The DNS Server was configured in the Johor Server Farm using Cisco Packet Tracer's server device. The following steps were taken:

1. A **static IPv4 address** was assigned to the DNS server according to the addressing plan (198.51.100.8).

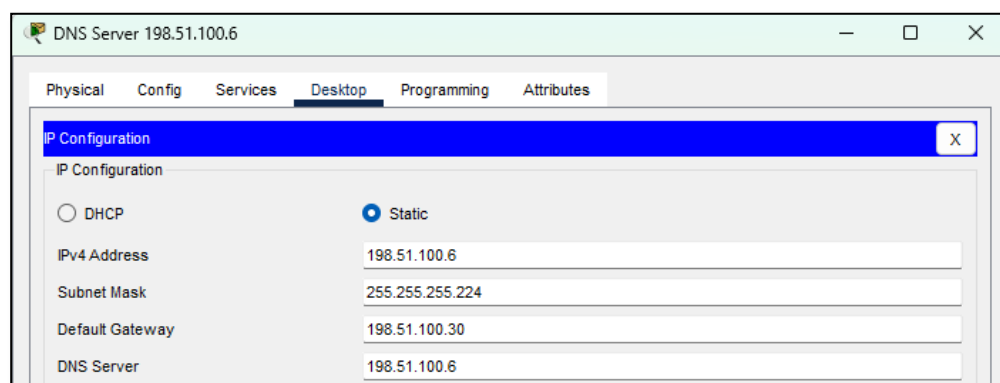


Figure 7.3.1.2 DNS Server IP Configuration

2. Within the server's **Services tab**, the **DNS service** was enabled.

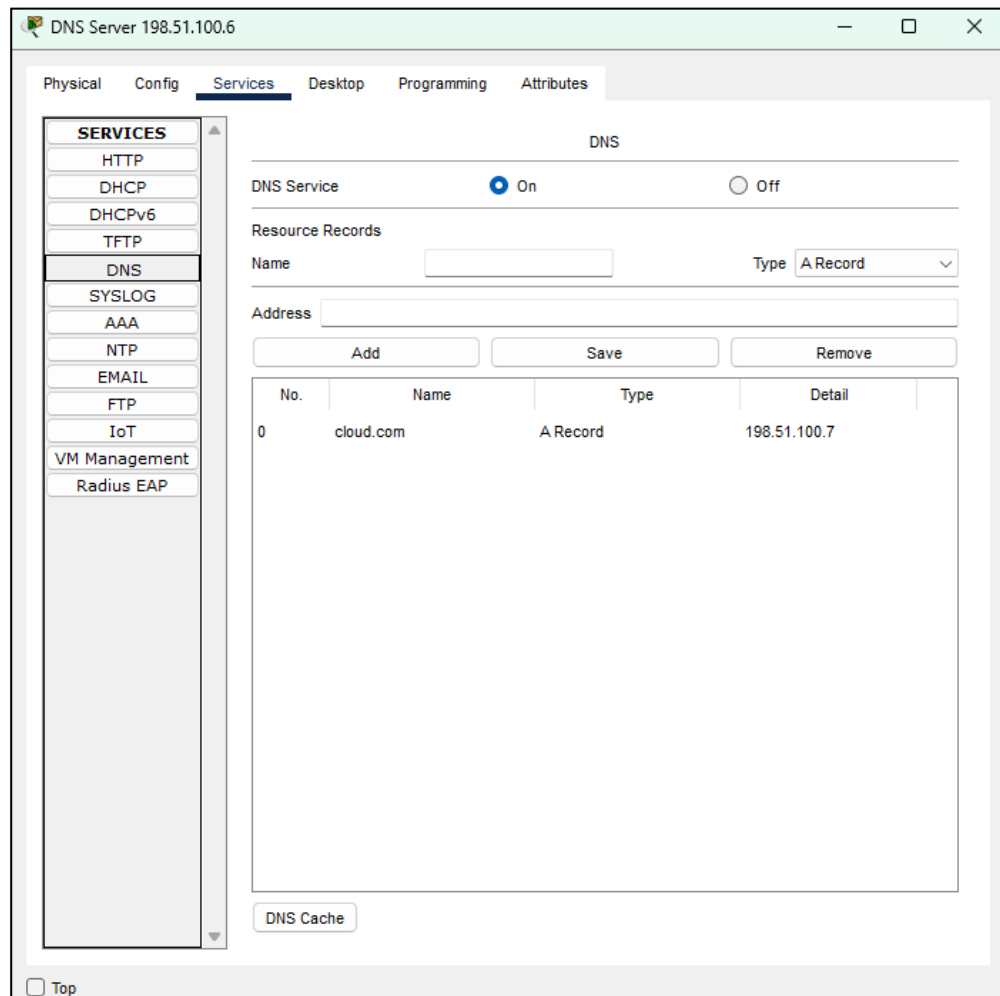


Figure 7.3.1.3 DNS Server Services View

3. **Resource records (A records)** were created to map hostnames to IP addresses, such as, `www.cloud.com` which would lead to our Web Server with the IP (198.51.100.7).
4. The configuration was saved to ensure persistence across simulation sessions.

7.3.2 Verification of DNS Server Configuration

To verify the success of the DNS Server configuration, go to any PC in the network, and go to the Web Browser within the Desktop of the PC.

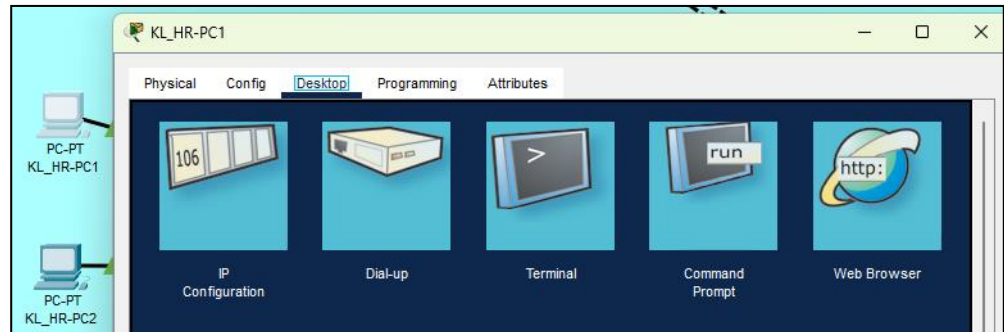


Figure 7.3.2.1 Desktop View of a computer

After opening the Web Browser, instead of entering the IP address, we will enter the DNS name (cloud.com) to try and access the Web Server configured in the network.



Figure 7.3.2.2 cloud.com Web Page

The output of using the domain name instead of the IP shows the same output, proving the configuration of the DNS server was successful.

7.4 DHCP Server

The DHCP Server in Johor Server Farm provides the organization with a centralized service for automatic IP address allocation, subnet mask, default gateway, and DNS information. This method is essential for reducing configuration errors, seamless network management, and smooth connectivity for all users.

7.4.1 Configuration Process

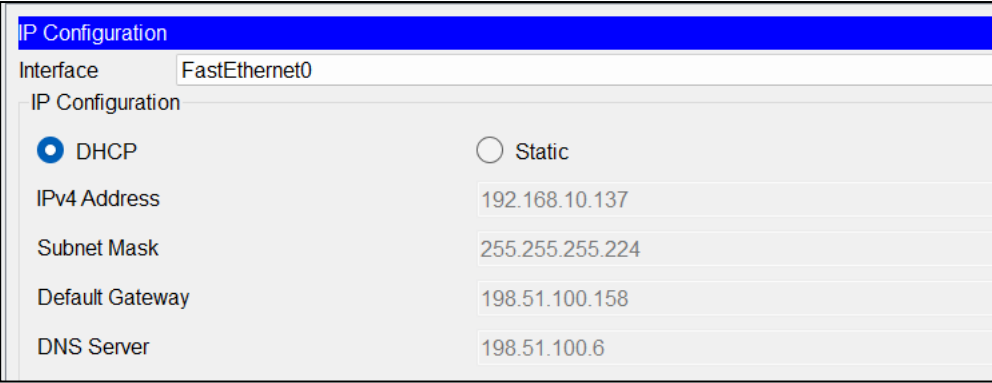
The DHCP Server has been placed in the Johor Server Farm to automatically assign IP configuration to end devices. By enabling the DHCP services, the server is configured with the department name, default gateway, IP address, subnet mask, and maximum number of users to connect efficiently without manual configuration.

The screenshot shows the DHCP Server configuration interface for IP 198.51.100.8. The 'Services' tab is active, displaying the DHCP configuration for the 'FastEthernet0' interface. The service is enabled (On). The configuration includes a pool named 'serverPool' with a default gateway of 0.0.0.0, DNS server of 0.0.0.0, and a maximum number of users of 512. Below the configuration fields, there is a table listing several DHCP pools.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
PG_IT	172.16.10.129	198.51.100.6	172.16.10.139	255.255.255...	21	0.0.0.0	172.16.10.68
PG_Mgmt	172.16.10.65	198.51.100.6	172.16.10.75	255.255.255...	50	0.0.0.0	172.16.10.68
PG_Delivery	172.16.10.1	198.51.100.6	172.16.10.10	255.255.255...	54	0.0.0.0	172.16.10.68
JB_Fin	10.10.1.62	198.51.100.6	10.10.1.7	255.255.255...	56	0.0.0.0	0.0.0.0
JB_VoiceVlan	10.10.1.142	198.51.100.6	10.10.1.135	255.255.255...	6	10.10.1.129	0.0.0.0
JB_R&D	10.10.1.126	198.51.100.6	10.10.1.67	255.255.255...	58	0.0.0.0	0.0.0.0
KL_Design	198.51.100.158	198.51.100.6	192.168.10.131	255.255.255...	26	0.0.0.0	0.0.0.0
KL_HR	198.51.100.126	198.51.100.6	192.168.10.67	255.255.255...	58	0.0.0.0	0.0.0.0

7.4.2 Verification of DHCP Server Configuration

After adding the correct information during the DHCP configuration process, it will be easier for the end devices to obtain valid IP addresses automatically. This method reduces manual configuration and ensures each device is assigned with the proper network setting.

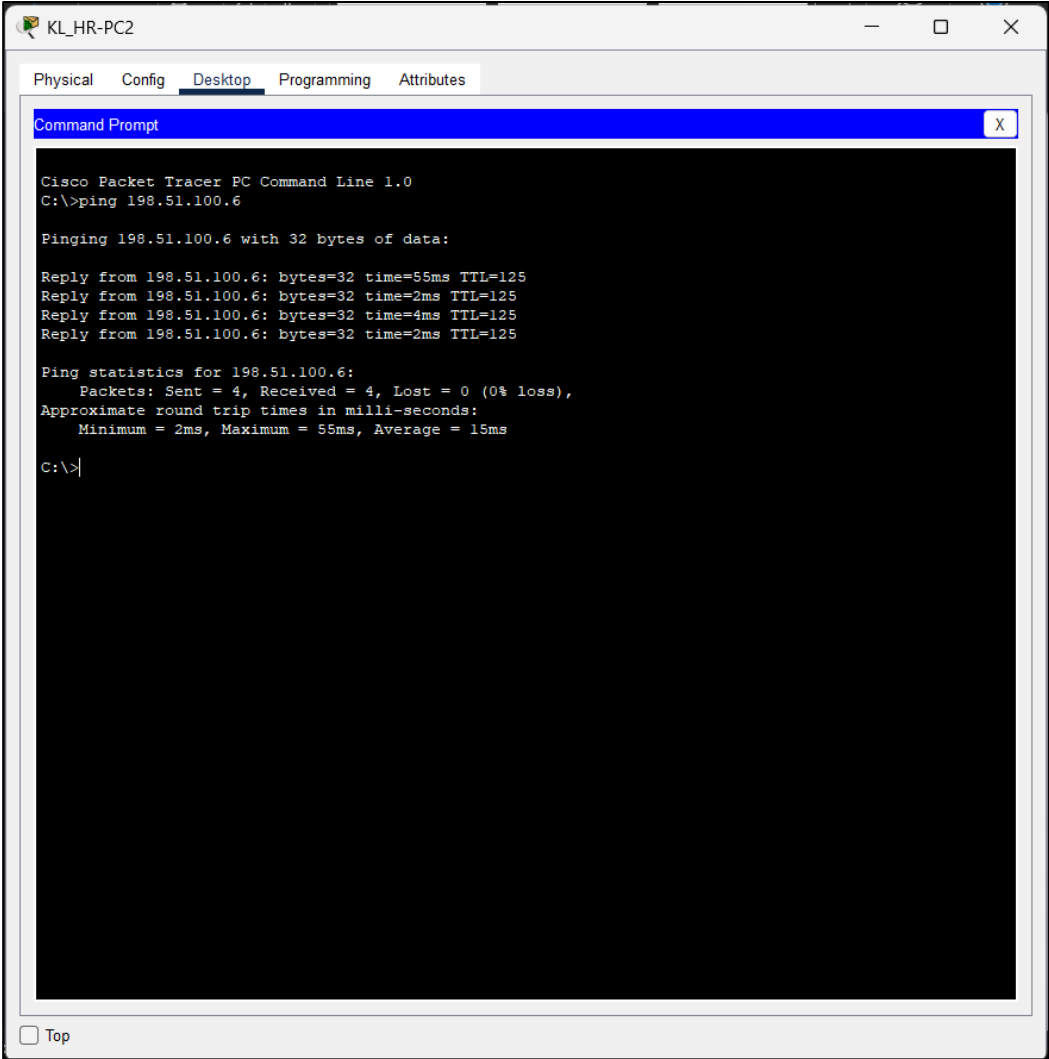


The screenshot displays the 'IP Configuration' window for the 'FastEthernet0' interface. The 'IP Configuration' section is expanded, showing the 'DHCP' option selected with a radio button. The 'Static' option is also visible but unselected. Below the radio buttons, the following fields are populated:

Field	Value
IPv4 Address	192.168.10.137
Subnet Mask	255.255.255.224
Default Gateway	198.51.100.158
DNS Server	198.51.100.6

8.0 Tracert and Ping Result

8.1 Connection from KL to Server Farm



```
KL_HR-PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 198.51.100.6

Pinging 198.51.100.6 with 32 bytes of data:

Reply from 198.51.100.6: bytes=32 time=55ms TTL=125
Reply from 198.51.100.6: bytes=32 time=2ms TTL=125
Reply from 198.51.100.6: bytes=32 time=4ms TTL=125
Reply from 198.51.100.6: bytes=32 time=2ms TTL=125

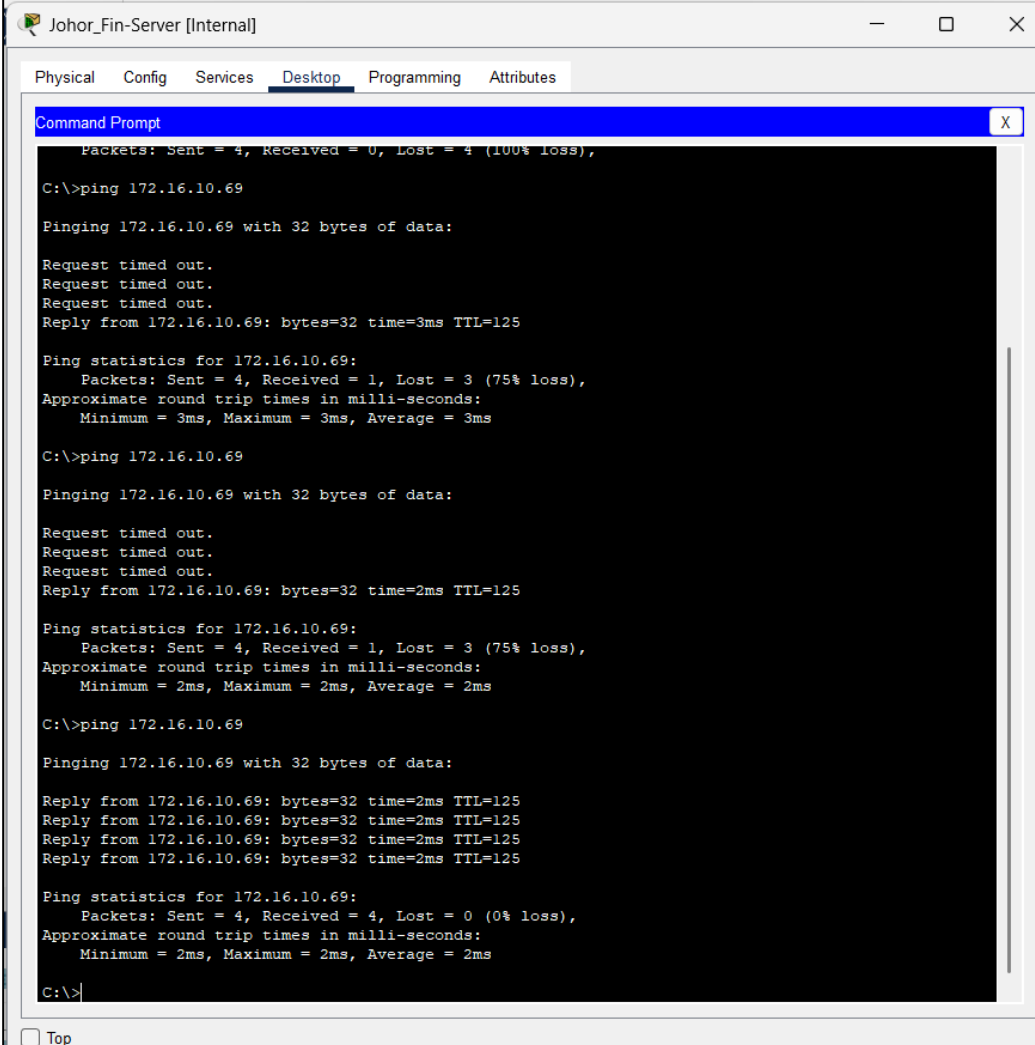
Ping statistics for 198.51.100.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 55ms, Average = 15ms

C:\>
```

Source: KL_HR-PC2 (192.168.10.68)

Destination: DNS Server (198.51.100.6)

8.2 Connection from Johor to Penang



```

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 172.16.10.69

Pinging 172.16.10.69 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Reply from 172.16.10.69: bytes=32 time=3ms TTL=125

Ping statistics for 172.16.10.69:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 3ms, Average = 3ms

C:\>ping 172.16.10.69

Pinging 172.16.10.69 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Reply from 172.16.10.69: bytes=32 time=2ms TTL=125

Ping statistics for 172.16.10.69:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\>ping 172.16.10.69

Pinging 172.16.10.69 with 32 bytes of data:

Reply from 172.16.10.69: bytes=32 time=2ms TTL=125
Reply from 172.16.10.69: bytes=32 time=2ms TTL=125
Reply from 172.16.10.69: bytes=32 time=2ms TTL=125
Reply from 172.16.10.69: bytes=32 time=2ms TTL=125

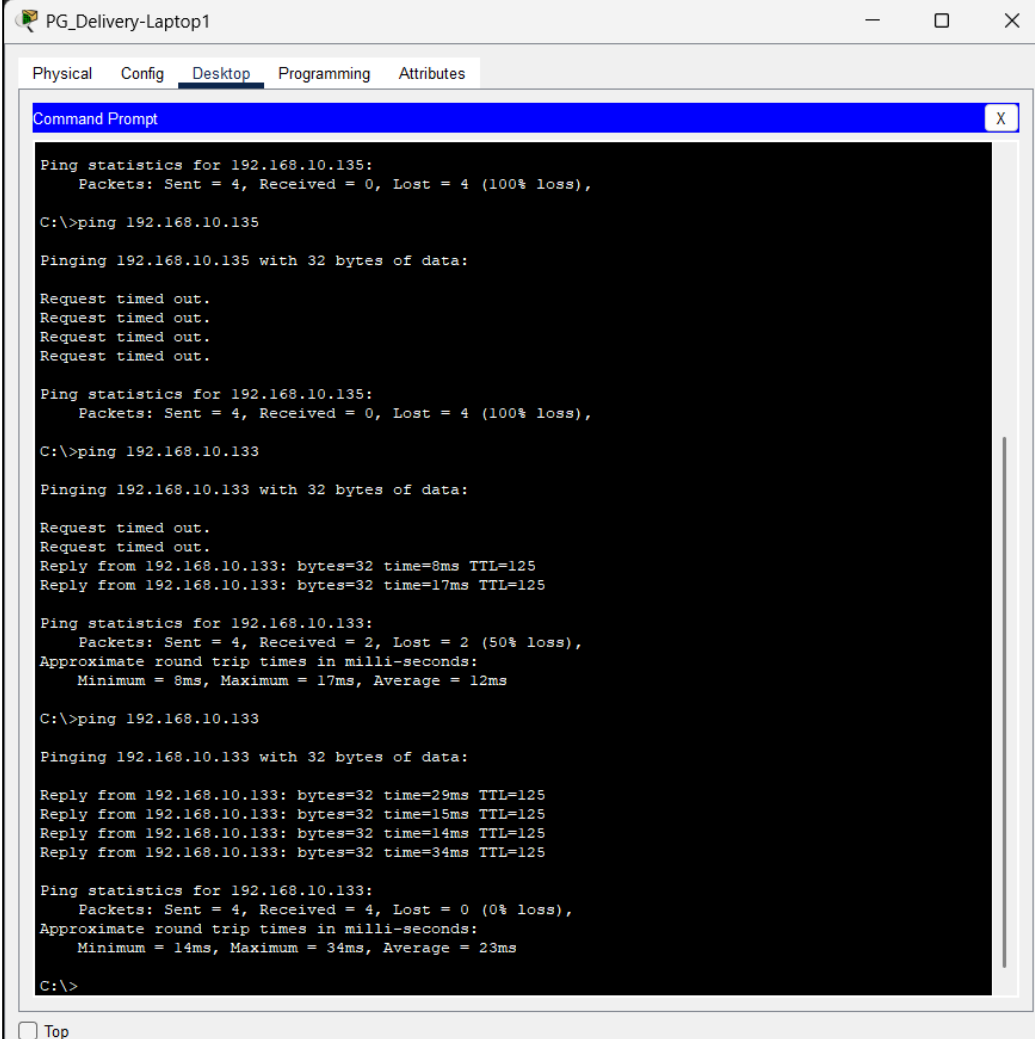
Ping statistics for 172.16.10.69:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\>|
```

Source: Johor_Fin-Server [Internal] (10.10.1.5)

Destination: PG_Mgmt-PC (172.16.10.69)

8.3 Connection from Penang to KL



The screenshot shows a Command Prompt window titled "PG_Delivery-Laptop1" with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window with the following text:

```
Command Prompt
Ping statistics for 192.168.10.135:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.135

Pinging 192.168.10.135 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.135:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.133

Pinging 192.168.10.133 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.10.133: bytes=32 time=8ms TTL=125
Reply from 192.168.10.133: bytes=32 time=17ms TTL=125

Ping statistics for 192.168.10.133:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 17ms, Average = 12ms

C:\>ping 192.168.10.133

Pinging 192.168.10.133 with 32 bytes of data:

Reply from 192.168.10.133: bytes=32 time=29ms TTL=125
Reply from 192.168.10.133: bytes=32 time=15ms TTL=125
Reply from 192.168.10.133: bytes=32 time=14ms TTL=125
Reply from 192.168.10.133: bytes=32 time=34ms TTL=125

Ping statistics for 192.168.10.133:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 34ms, Average = 23ms

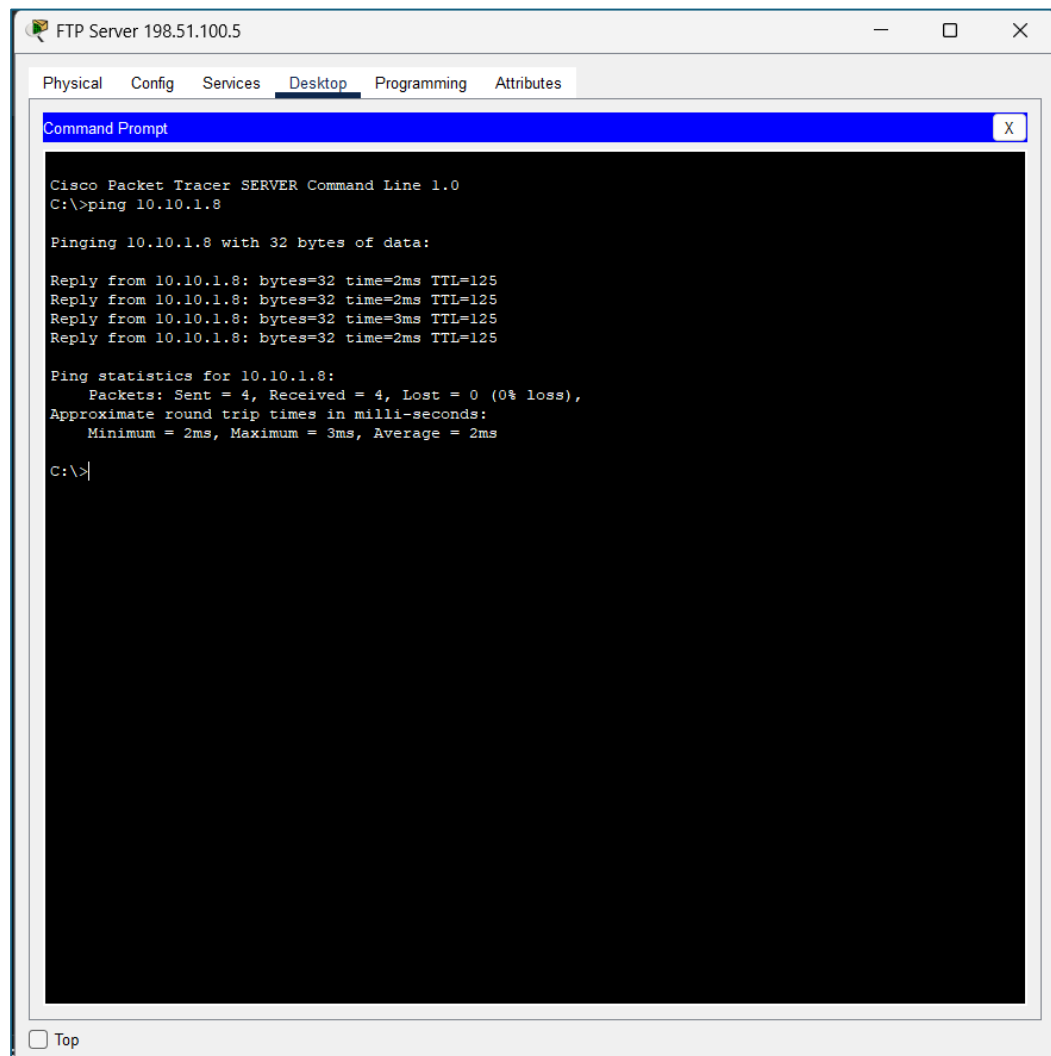
C:\>
```

At the bottom left of the Command Prompt window, there is a checkbox labeled "Top".

Source: PG_Delivery-Laptop1(172.16.10.20)

Destination: KL_Design-PC3 (192.168.10.133)

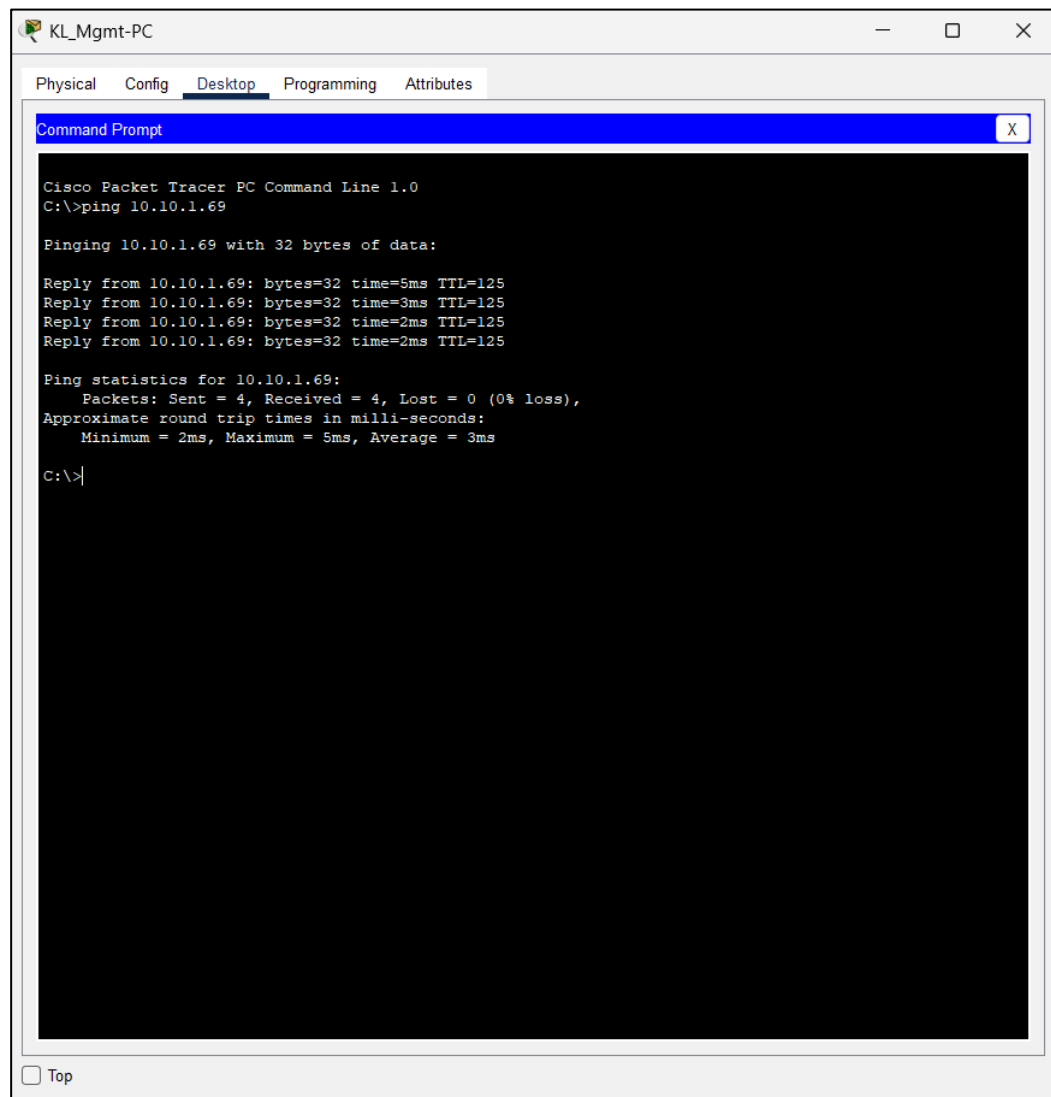
8.4 Connection from Server Farm to Johor



Source: FTP Server (198.51.100.5)

Destination: Johor_Fin-PC (10.10.1.8)

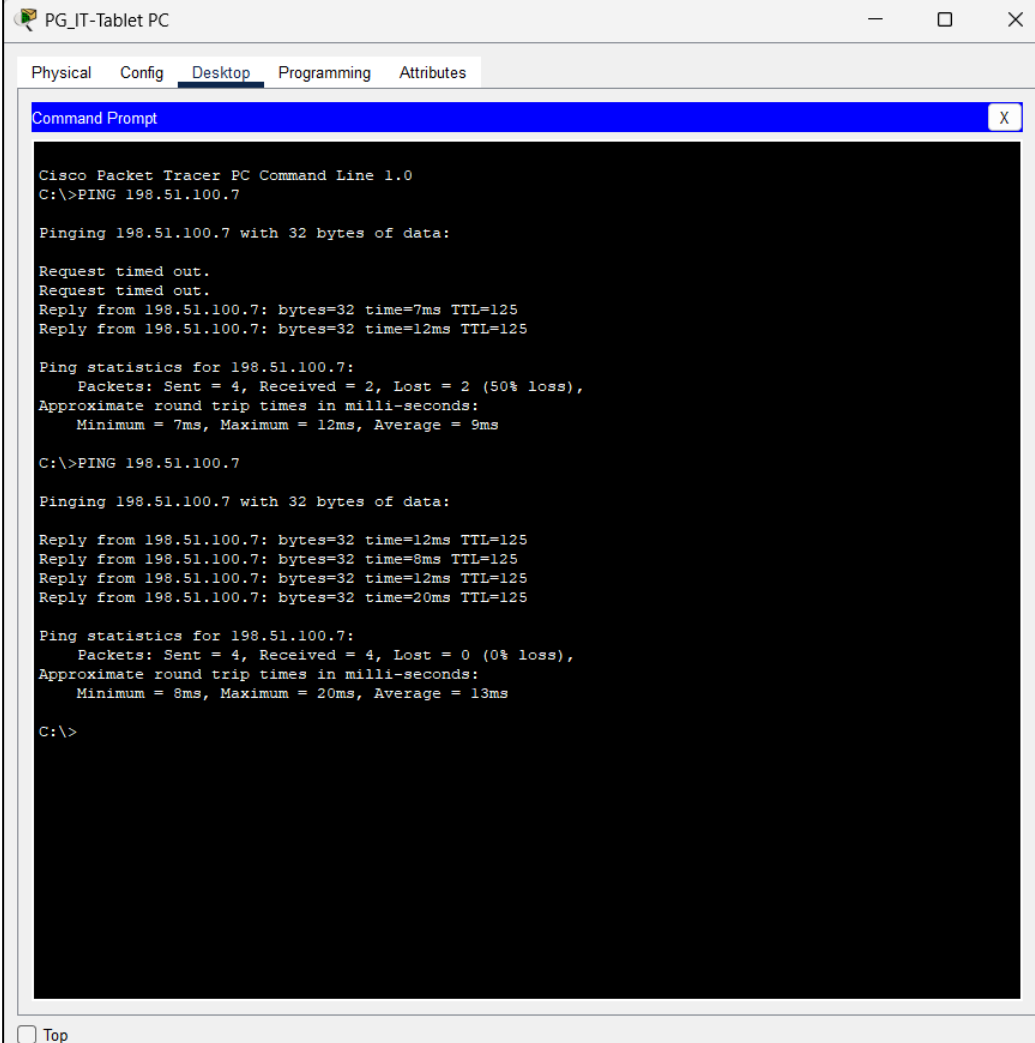
8.5 Connection from KL to Johor



Source: KL_Mgmt-PC (192.168.10.8)

Destination: JB_R&D-PC2 (10.10.1.69)

8.6 Connection from Penang to Server Farm



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named PG_IT-Tablet PC. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>PING 198.51.100.7

Pinging 198.51.100.7 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 198.51.100.7: bytes=32 time=7ms TTL=125
Reply from 198.51.100.7: bytes=32 time=12ms TTL=125

Ping statistics for 198.51.100.7:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 12ms, Average = 9ms

C:\>PING 198.51.100.7

Pinging 198.51.100.7 with 32 bytes of data:

Reply from 198.51.100.7: bytes=32 time=12ms TTL=125
Reply from 198.51.100.7: bytes=32 time=8ms TTL=125
Reply from 198.51.100.7: bytes=32 time=12ms TTL=125
Reply from 198.51.100.7: bytes=32 time=20ms TTL=125

Ping statistics for 198.51.100.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 20ms, Average = 13ms

C:\>
```

At the bottom left of the window, there is a checkbox labeled "Top" which is currently unchecked.

Source: PG_IT-Tablet PC (172.16.10.149)

Destination: Web Server (198.51.100.7)

9.0 Basic Security Mechanisms

9.1 SSH

SSH allows us to manage and configure switches and routers securely without using the console cables. This allows network administrators to tightly control multiple switches and routers in remote locations. By implementing SSH allows all communication to be encrypted which prevents hackers to track on our command and password.

```
Unauthorized access is prohibited!

User Access Verification

Password:

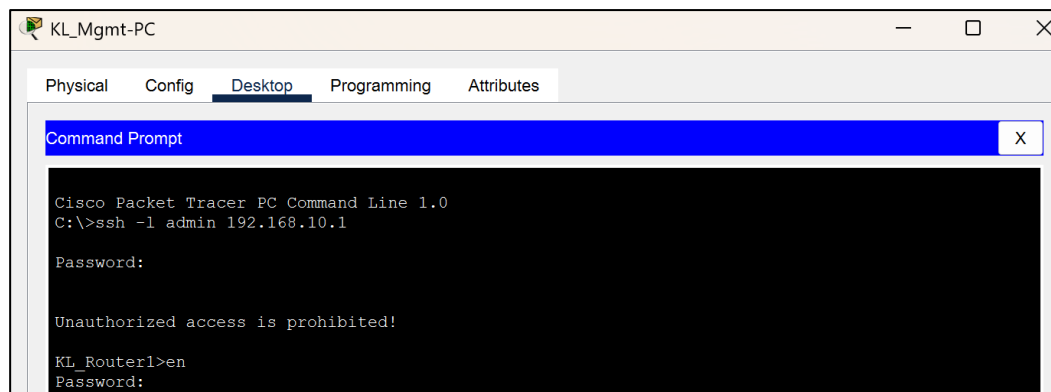
KL_Router2>en
Password:
KL_Router2#
```

```
KL_Mgmt_SW>en
KL_Mgmt_SW#config t
Enter configuration commands, one per line. End with CNTL/Z.
KL_Mgmt_SW(config)#ip domain-name kl.com
KL_Mgmt_SW(config)#crypto key generate rsa
The name for the keys will be: KL_Mgmt_SW.kl.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

KL_Mgmt_SW(config)#ip ssh ver 2
*Mar 1 0:52:11.75: %SSH-5-ENABLED: SSH 1.99 has been enabled
KL_Mgmt_SW(config)#username admin secret ssh
KL_Mgmt_SW(config)#enable secret cisco
KL_Mgmt_SW(config)#line vty 0 4
KL_Mgmt_SW(config-line)#transport input ssh
KL_Mgmt_SW(config-line)#login local
KL_Mgmt_SW(config-line)#exit
KL_Mgmt_SW(config)#line console 0
KL_Mgmt_SW(config-line)#password cisco
KL_Mgmt_SW(config-line)#login
KL_Mgmt_SW(config-line)#exit
KL_Mgmt_SW(config)#banner motd #
Enter TEXT message. End with the character '#'.
Unauthorized access is prohibited! #

KL_Mgmt_SW(config)#do wr
Building configuration...
[OK]
```



Test SSH connectivity from another device (PC)

9.2 Layer 2 Attacks and Mitigation Techniques

9.2.1 First Layer 2 Attack

9.2.1.1 Layer 2 Attack - MAC Flooding Attack

A MAC flooding attack is another Layer 2 attack that exploits the behavior of network switches by overloading their MAC address table with fake addresses. This forces the switch to send traffic to all ports instead of the correct destination. As a result, attackers take this chance to intercept data and disrupt the network, causing slower performance, unauthorized access, and potential data breaches for businesses.

9.2.1.2 Mitigation

Port Security Mitigation

Port security is one of the most effective ways to mitigate MAC address table flooding attacks on the network switches by limiting the number of MAC addresses that can be learned on each port.

9.2.2 Second Layer 2 Attack

9.2.3 Third Layer 2 Attack

9.2.3.1 Layer 2 Attack – STP Attack

Spanning Tree Protocol (STP) is a Layer 2 mechanism used in switched networks to prevent loops and ensure a stable forwarding topology. While STP is important, it also introduces a potential attack surface because switch will implicitly trust any Bridge Protocol Data Unit (BPDU) they receive. Attackers can exploit this trust by sending specially crafted malicious BPDUs into the network. Such attacks are called STP attacks which are aimed to disrupt the topology, intercept traffic, create instability within the switching infrastructure.

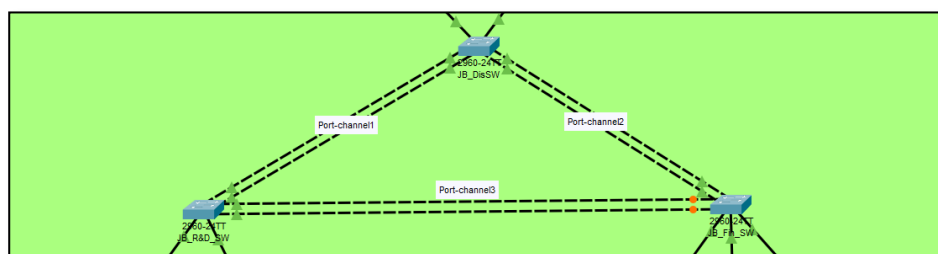


Figure 9.2.3.1.1 Spanning Tree Topology

BPDU Flooding Attack:

A BPDU flooding attack involves an attacker generating a large number of fake or malicious BPDUs to overload the STP process. By continuously forcing switches to recalculate the spanning tree, this attack creates frequent topology changes, causing high CPU usage, packet loss, and service disruption. The constant reconvergence prevents the network from stabilizing and can lead to denial of service, as legitimate traffic experiences delays or is dropped during repeated recalculations.

Root Bridge Attack (STP Takeover):

In a root bridge attack, an attacker sends falsified BPDUs advertising an artificially low bridge priority to trick legitimate switches into electing the attacker's device as the root bridge. Once this manipulation succeeds, the attacker's switch becomes the central forwarding point in the topology, forcing network traffic to pass through it. This allows the attacker to intercept, analyze, or even alter data flows, while also creating instability in the topology if the attacker disconnects or misconfigures their device.

9.2.3.2 Mitigation Methods

BPDU Flooding Attack Mitigation

To defend against BPDU flooding, we applied PortFast and BPDU Guard on all access ports of the management, HR, design, R&D, and finance switches in KL and Johor branch. PortFast ensures that end devices connect quickly without participating in STP convergence, while BPDU Guard shuts down the port immediately if any BPDU is received from an unauthorized device.

```
KL_Design_SW>en
KL_Design_SW#config t
Enter configuration commands, one per line. End with CNTL/Z.
KL_Design_SW(config)#int ra fa0/11-13
KL_Design_SW(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/11 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/12 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/13 but will only
have effect when the interface is in a non-trunking mode.
KL_Design_SW(config-if-range)#spanning-tree bpduguard enable
```

Figure 9.2.3.2.1 Configuration of PortFast

Root Bridge Attack Mitigation

In our Packet Tracer topology, we prevented by explicitly configuring the intended distribution switches in our topology, KL_DisSw and JB_DisSw as the root primary for their VLANs. This ensures they always win the root election and prevent Root Bridge Attack.

```
JB_DisSW>en
JB_DisSW#config t
Enter configuration commands, one per line. End with CNTL/Z.
JB_DisSW(config)#spanning-tree ?
    mode          Spanning tree operating mode
    portfast       Spanning tree portfast options
    vlan           VLAN Switch Spanning Tree
JB_DisSW(config)#spanning-tree vlan 40,50,60,99 root primary
JB_DisSW(config)#
```

Figure 9.2.3.2.2 Configuration of root primary

9.2.4 Fourth Layer 2 Attack

9.2.4.1 Layer 2 Attack – Unused Port Attack

An unused port attack is a network attack that occurs when an attacker plugs an unauthorized device into an available switch access port. For example, in an unlocked wiring closet or on a desk in the company. Because many access ports are left in their default, permissive state, this can grant the attacker immediate network connectivity and potentially access to internal resources, pivot across VLANs, or run reconnaissance and exploitation tools. These attacks exploit lax physical security and default port configuration.

Device Name: KL_DisSW				
Custom Device Model: 2960 IOS15				
Hostname: KL_DisSW				
Port	Link	VLAN	IP Address	MAC Address
Port-channel1	Up	--	<not set>	000B.BECB.4C16
Port-channel2	Up	--	<not set>	0003.E43E.A66A
Port-channel3	Up	--	<not set>	0010.11B6.0204
FastEthernet0/1	Up	--	--	0002.1612.B101
FastEthernet0/2	Up	--	--	0002.1612.B102
FastEthernet0/3	Up	--	--	0002.1612.B103
FastEthernet0/4	Up	--	--	0002.1612.B104
FastEthernet0/5	Up	--	--	0002.1612.B105
FastEthernet0/6	Up	--	--	0002.1612.B106
FastEthernet0/7	Down	100	--	0002.1612.B107
FastEthernet0/8	Down	100	--	0002.1612.B108
FastEthernet0/9	Down	100	--	0002.1612.B109
FastEthernet0/10	Down	100	--	0002.1612.B10A
FastEthernet0/11	Down	100	--	0002.1612.B10B
FastEthernet0/12	Down	100	--	0002.1612.B10C
FastEthernet0/13	Down	100	--	0002.1612.B10D
FastEthernet0/14	Down	100	--	0002.1612.B10E
FastEthernet0/15	Down	100	--	0002.1612.B10F
FastEthernet0/16	Down	100	--	0002.1612.B110
FastEthernet0/17	Down	100	--	0002.1612.B111
FastEthernet0/18	Down	100	--	0002.1612.B112
FastEthernet0/19	Down	100	--	0002.1612.B113
FastEthernet0/20	Down	100	--	0002.1612.B114
FastEthernet0/21	Down	100	--	0002.1612.B115
FastEthernet0/22	Down	100	--	0002.1612.B116
FastEthernet0/23	Down	100	--	0002.1612.B117
FastEthernet0/24	Down	100	--	0002.1612.B118
GigabitEthernet0/1	Up	--	--	0002.1612.B119
GigabitEthernet0/2	Up	--	--	0002.1612.B11A
Vlan1	Down	1	<not set>	0001.4362.D2C3
Vlan99	Up	99	192.168.10.167/28	0001.4362.D201
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > KL_DisSW				

Figure 9.2.4.1.1 All ports Status View of KL Switch

9.2.4.2 Mitigation Methods

Create Blackhole

To prevent attackers from flooding the network with spam packets through unused ports, a blackhole VLAN is created. A BlackHole VLAN is a special, unused VLAN created specifically to isolate switch ports that should not carry any production traffic. When unused ports are placed in this VLAN, any device that is accidentally or maliciously connected to those ports will be logically isolated from the rest of the network. Because the BlackHole VLAN has no Layer 3 interface, no routing, and no access to servers or user VLANs, devices in it cannot communicate beyond their own segment.

Shutdown Unused Ports

One method to prevent unused port attacks is to shut down the ports completely. By shutting down idle ports, we are able to eliminate the risk of unauthorized devices gaining access through unused interfaces. This ensures that only designated active ports participate in the production network, preventing any unauthorized access through the ports.

For our topology, we set VLAN 100 as the BlackHole VLAN for every branch's network. In addition to that we also make sure to shut down any and all unused ports. By taking these precautions, we are able to greatly mitigate the risk of attacks happening on our networks coming from unused ports.

10.0 Recommendations to Improve the Security Mechanisms

To further improve the security posture of the organization, the additional mechanisms need to be embraced besides the current configurations. The following are suggested to enhance the defensive as well as the survivability of the network infrastructure.

The initial solution is implementation of **honey pot** system. A honey pot refers to a specifically isolated network that makes itself very appealing to attackers so that the attackers do not get access to the production network. The malicious actors disclose their tactics, techniques, and procedures by communicating with the honey pot. This enables the security team to gather intelligence, research on how attacks are undertaken and optimize the defences all without compromising actual systems.

Besides that, the second suggestion is putting in place a **bastion host**. A bastion host is a hardened server located at the network boundary, and it usually in a demilitarized zone (DMZ). Its primary use is to act as a regulated access point to administrative or sensitive access. This approach will mean that the external connections pass through this safe host, thereby minimizing the total attack area of the organization, consolidating the authentication process, and enhancing accountability of the user activities.

Furthermore, the third suggestion is implementation of **network segmentation**. Segmentation separates internal network into small and isolated segments in terms of departments or functions. This will stop the movement of attackers horizontally through systems when they have already compromised a single segment. It also enhances visibility to track traffic and allow the deployment of more specific security policies against important resources.

As a nutshell, the implementation of honey pot, bastion hosts, and network segmentation would offer the organization a particular and resilient defence approach. Such mechanisms supplement the existing protection mechanisms, reduce the risks of internal and external attacks, and secure the continuity of the safe business processes.

11.0 Conclusion

Aaa

12.0 References

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